María José Cuéllar Mejía

Implementation of State of Charge and State of Health Algorithm for a Battery Management System for a Racing Electric Motorcycle

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Universitat Politècnica de Catalunya (UPC)
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Supervision

Prof. Dr. Francisco Díaz González
Prof. Dr. Joan Gabriel Bergas Jané

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Abstract

Electric racing motorcycles require accurate, real-time knowledge of the battery's State of Charge (SOC) and State of Health (SOH) to guarantee safe and reliable operation under the highly dynamic current profiles typical of a race. This work develops and validates a model-based estimation framework for the simultaneous estimation of SOC and SOH of the WeLion SHP270-30 Nickel-Manganese-Cobalt (NMC) lithium-ion cell (30 Ah nominal capacity) used in the eRacing electric motorcycle developed at ETSEIB.

The battery was represented using a second-order Thévenin equivalent circuit model (ECM), consisting of a voltage source, a series resistance, and two parallel resistor–capacitor branches. The cell was first characterised experimentally through discharge, charge, and pulse-discharge tests, from which the actual capacity and the Open Circuit Voltage-SOC relationship were obtained. The equivalent-circuit parameters were subsequently identified using an algebraic method, applied to a two-level step-current test repeated across different SOC operating points. Finally, two Extended Kalman Filters (EKF) were implemented in MATLAB, a SOC filter, and a dual EKF that simultaneously estimates SOC and the SOH parameters, the series resistance R_0 and the capacity Q.

Both filters were validated using experimental data from three current profiles: a charge cycle, a cyclic pulse discharge, and a random step-current discharge simulating race conditions. The SOC filter achieved root-mean-square errors (RMSE) between 0.69% and 1.14% relative to a coulomb-counting reference, remaining within the estimated 3σ uncertainty bounds throughout all tests. The dual EKF achieved comparable SOC accuracy (RMSE between 0.94% and 1.39%) while simultaneously obtaining the capacity and resistance estimates to values consistent with the characterised cell.

The results confirm that a second-order Thévenin ECM combined with a dual EKF provides a computationally efficient and sufficiently accurate framework for real-time SOC and SOH estimation under the dynamic current profiles found in motorsport applications. The algebraic parameter identification method proved effective across most of the operating range, although reduced reliability was observed at the extremes of the state of charge, consistent with the known numerical sensitivity of the method and the limitations of the equivalent-circuit model in capturing the underlying electrochemical behaviour in these regions. Future work should extend the framework to account for temperature dependence and validate its performance under real race data.

Keywords: State of Charge, State of Health, Extended Kalman Filter, Equivalent-Circuit Model, lithium-ion battery, battery characterisation.

Resum

Les motocicletes elèctriques de competició requereixen un coneixement precís i en temps real de l'Estat de Càrrega (SOC) i l'Estat de Salut (SOH) de la bateria per garantir un funcionament segur i fiable sota els perfils de corrent altament dinàmics propis d'una cursa. Aquest treball desenvolupa i valida un marc d'estimació basat en un model per a l'estimació simultània del SOC i el SOH de la cel·la de liti-ió WeLion SHP270–30, de química Níquel-Manganès-Cobalt (NMC) i capacitat nominal de 30 Ah, utilitzada a la motocicleta elèctrica eRacing desenvolupada a l'ETSEIB.
La bateria es va representar mitjançant un model de circuit equivalent (ECM) de Thévenin de segon ordre, format per una font de tensió, una resistència en sèrie i dues branques paral·leles de resistència-condensador. La cel·la es va caracteritzar experimentalment mitjançant assajos de descàrrega contínua, càrrega i descàrrega per polsos, dels quals es van obtenir la capacitat real i la relació OCV–SOC. Posteriorment, els paràmetres del circuit equivalent es van identificar mitjançant un mètode algebraic, aplicat a un assaig d'esglaó de corrent de dos nivells, repetit en diferents punts d'operació del SOC. Finalment, es van implementar dos Filtres de Kalman Estesos (EKF) en MATLAB, un filtre de SOC, i un EKF dual que estima simultàniament el SOC i els paràmetres del SOH, la resistència en sèrie R_0 i la capacitat Q.
Ambdós filtres es van validar amb dades experimentals de tres perfils de corrent: un cicle de càrrega, una descàrrega cíclica per polsos, i una descàrrega d'esglaons de corrent aleatoris que simulen les condicions d'una cursa. El filtre de SOC va assolir errors quadràtics mitjans (RMSE) d'entre el $0,69\%$ i l'$1,14\%$ respecte a una referència de comptatge de coulombs, mantenint-se dins dels límits d'incertesa de 3σ estimats en tots els assajos. L'EKF dual va assolir una precisió del SOC comparable (RMSE d'entre el $0,94\%$ i l'$1,39\%$) mentre convergia simultàniament les estimacions de capacitat i resistència a valors consistents amb la cel·la caracteritzada.
Els resultats confirmen que un ECM de Thévenin de segon ordre combinat amb un EKF dual proporciona un marc computacionalment eficient i prou precís per l'estimació en temps real del SOC i el SOH sota els perfils de corrent dinàmics propis de les aplicacions d'automobilisme esportiu. El mètode algebraic d'identificació de paràmetres es va mostrar efectiu en la major part del rang d'operació, tot i que es va observar una fiabilitat reduïda als extrems de l'estat de càrrega, coherent amb la sensibilitat numèrica coneguda del mètode i amb les limitacions del model de circuit equivalent per capturar el comportament electroquímic subjacent en aquestes regions. Els treballs futurs haurien d'ampliar el marc per tenir en compte la dependència de la temperatura i validar-ne el rendiment amb dades reals de telemetria d'una cursa.

Paraules clau: Estat de Càrrega, Estat de Salut, Filtre de Kalman Estès, Model de Circuit Equivalent, bateria de liti-ió, caracterització de bateries.

Resumen

Las motocicletas eléctricas de competencia requieren un conocimiento del Estado de Carga (SOC) preciso y en tiempo real, así como el Estado de Salud (SOH) de la batería para poder garantizar un funcionamiento seguro y confiable bajo perfiles dinámicos de corriente correspondientes a los de una carrera. Este trabajo desarrolla y valida una metodología para la estimación simultánea del SOC y el SOH de la celda de litio-ion WeLion SHP270–30, de química Níquel-Manganeso-Cobalto (NMC), con una capacidad nominal de $30Ah$, la cual es la correspondiente a la utilizada en la motocicleta eléctrica eRacing desarrollada en ETSEIB.
La batería se representó mediante un modelo de circuito equivalente (ECM) de Thévenin de segundo orden, conformado por una fuente de voltaje, una resistencia en serie y dos ramas con una resistencia y un capacitor en paralelo. La celda se caracterizó experimentalmente mediante pruebas de descarga, carga y descarga por pulsos, de las cuales se obtuvieron la capacidad real y la relación OCV–SOC. Posteriormente, los parámetros del circuito equivalente se identificaron mediante un método algebraico, aplicado a una prueba conformada por dos escalones de corriente y repetida en distintos puntos de operación del SOC. Finalmente, se implementaron dos filtros de Kalman Extendidos (EKF) en MATLAB, un filtro de SOC, y un dual EKF que estima simultáneamente el SOC y los parámetros del SOH, siendo estos parámetros la resistencia en serie R_0 y la capacidad Q.
Ambos filtros se validaron con datos experimentales obtenidos de tres perfiles de corriente: un ciclo de carga, uno de descarga cíclica por pulsos, y uno de descarga de escalones de corriente aleatorios que simulan las condiciones de una carrera. El filtro de SOC alcanzó errores (RMSE) de entre 0.69% y 1.14% respecto a la referencia de Coulomb Counting, manteniéndose dentro de los límites de incertidumbre de 3σ estimados en todas las pruebas. El EKF dual alcanzó una precisión del SOC comparable (RMSE de entre 0.94% y 1.39%) mientras obtenía al mismo tiempo las estimaciones de capacidad y resistencia a valores consistentes con la celda caracterizada.
Los resultados confirman que un ECM de Thévenin de segundo orden combinado con un dual EKF proporciona un marco computacionalmente eficiente y suficientemente preciso para la estimación en tiempo real del SOC y el SOH bajo perfiles de corriente dinámicos aproximadamente característicos de las aplicaciones del automovilismo deportivo. El método algebraico de identificación de parámetros demostró ser efectivo en la mayor parte del rango de operación, aunque se observó una confiabilidad reducida en los extremos del estado de carga, congruente con la sensibilidad numérica conocida del método y con las limitaciones del modelo de circuito equivalente para capturar el comportamiento electroquímico en estas regiones. Futuros trabajos deberían de ampliar el marco para considerar la dependencia con la temperatura y validar su desempeño con datos reales de telemetría de carrera.

Palabras clave: Estado de Carga, Estado de Salud, Filtro de Kalman Extendido, Modelo de Circuito Equivalente, batería de litio-ion, caracterización de baterías.

Contents

Abstract ii

Resum iii

Resumen iv

Nomenclature vii

1 State Of Art 5
- 1.0.1 Empirical Estimation Methods . . . 5
- 1.0.2 Battery Modelling Approaches . . . 5
- 1.0.3 Adaptive and Fuzzy Logic Methods for Estimation . . . 6
- 1.0.4 Data-Driven and Machine-Learning Models . . . 6
- 1.0.5 Kalman Filter Methods for SOC and SOH Estimation . . . 7
- 1.0.6 Contribution . . . 7

2 Methodology 8
- 2.1 Battery Model . . . 8
  - 2.1.1 Racing Motorcycle Battery Model . . . 9
- 2.2 OCV-SOC Curve Characterisation . . . 9
  - 2.2.1 Discharge Test . . . 10
  - 2.2.2 Charge Test . . . 11
  - 2.2.3 Pulses Test . . . 11
- 2.3 Parameter Identification . . . 13
  - 2.3.1 Parameter Identification - Test Setup . . . 13
  - 2.3.2 Test Steps . . . 14
- 2.4 Kalman Filter . . . 15
  - 2.4.1 Extended Kalman Filter . . . 16
- 2.5 Dual Estimation Technique . . . 16
- 2.6 EKF Implementation . . . 17
  - 2.6.1 EKF for SOC Estimation . . . 18
  - 2.6.2 EKF for SOH Estimation . . . 19
  - 2.6.3 EKF Dual Implementation for SOC and SOH Estimation . . . 22
  - 2.6.4 Recursive Algorithm . . . 22

3 Results 25
- 3.1 OCV-SOC Curve Characterisation . . . 25
  - 3.1.1 Discharge Test . . . 25
  - 3.1.2 Charge Test . . . 26
  - 3.1.3 Pulses Test . . . 27
- 3.2 Parameter Identification . . . 29
  - 3.2.1 R_0, R_1, R_2, C_1, C_2 parameters . . . 29
- 3.3 Extended Kalman Filter . . . 32

3.3.1 Estimation of SOC with EKF . . . 32
3.3.2 Estimation of SOC and SOH with Dual EKF . . . 35
3.3.3 Estimation Comparison between SOC & SOH EKF and SOC filter . . . 41

4 Planning **45**

5 Economic Assessment **47**
5.1 Labour Cost . . . 47
5.2 Operational Expenses . . . 47
5.3 Experimental Materials . . . 48
5.4 Software Licensing . . . 48
5.5 Total Cost . . . 48
5.6 Economic Assessment Discussion . . . 49

6 Environmental Assessment **50**
6.1 Electricity Consumption . . . 50
6.2 Greenhouse Gas Emissions . . . 51
6.3 Environmental Impact Discussion . . . 51

7 Social and Gender Equality Assessment **52**
7.1 Access to the Technology . . . 52
7.2 Differentiated Impacts of the Research . . . 52
7.3 Gender Distribution in Cited Sources . . . 52
7.4 Relevant Sustainable Development Goals . . . 53
7.5 Battery Characterisation . . . 54
7.6 Parameter Identification . . . 54
7.7 SOC Estimation with EKF . . . 55
7.8 Simultaneous SOC and SOH Estimation with Dual EKF . . . 55
7.9 Limitations and Future Work . . . 56

Nomenclature

Acronyms and Abbreviations

BMS	Battery Management System
BP	Backpropagation
CC	Coulomb Counting
CC-CV	Constant-current Constant-voltage
CO_2	Carbon Dioxide
CO_2e	Carbon Dioxide Equivalent
ECM	Equivalent Circuit Model
ECTS	European Credit Transfer and Accumulation System
EKF	Extended Kalman Filter
ESR	Equivalent Series Resistance
EV	Electric Vehicle
KF	Kalman Filter
Li-ion	Lithium-ion
Li-Po	Lithium Polymer
MILS	Multiple Innovation Least Squares
NMC	Nickel-Manganese-Cobalt
OCV	Open-Circuit Voltage
PBM	Physics-based models
RLS	Recursive Least Squares
RMSE	Root Mean Square Error
SDG	Sustainable Development Goals
SOC	State of Charge
SOH	State of Health
UKF	Unscented Kalman Filter

Preface

The transportation sector is the primary source of direct greenhouse gas emissions in the United States, accounting for 28% of the national total [19]. In the European Union, this sector is responsible for approximately one-fifth of total Carbon Dioxide (CO_2) emissions. The shift towards electric mobility is being accelerated by regulations, such as the EU's plan for all new cars and vans to be zero-emission by 2035. Consequently, the EU estimates that battery-electric vehicles will dominate the market [6].

Electric Vehicles (EVs) are constrained by their battery energy capacity, as the size and energy density of the battery pack determine the vehicle's driving range [17]. Lithium-ion (Li-ion) batteries have played a major role in the decarbonisation of mobility due to their lightweight characteristics and high efficiency. Li-ion batteries have addressed these limitations and have emerged as a widespread technology for electric mobility [9], replacing and outperforming other battery technologies such as lead-acid or nickel-metal hydride [18]. However, their performance depends on factors such as the temperature, current loads, and charge-discharge cycles [4]. Another critical aspect concerns the safety risks associated with overcharging or overdischarging, as these can trigger internal chemical reactions or short-circuiting [17]. Due to these challenges and to guarantee optimal performance, longevity, and safety, accurate monitoring of the battery state, as well as individual cell voltages and temperatures, is essential [4].

Among critical internal states required for effective monitoring are the State of Charge (SOC) and State of Health (SOH). As these variables cannot be directly measured by any sensor, they must be estimated from measurable signals, such as voltage, current, and temperature. Estimation algorithms vary in complexity and accuracy. In high-performance and safety-critical systems, the consequences of estimation errors extend beyond suboptimal performance and may constitute a safety risk. Inaccurate SOC estimation can lead to an inefficient use of battery capacity and cell damage. Similarly, incorrect SOH can lead to unpredictable system behaviour and unreliability [17].

The eRacing team at ETSEIB designs and develops an electric motorcycle to compete at the international MotoStudent competition. To meet the performance requirements at different track conditions [15], the vehicle is powered by Nickel-Manganese-Cobalt (NMC) batteries, a type of Li-ion battery chemistry. The system operates at high voltages, which requires safety protocols to mitigate operational hazards. Consequently, the Battery Management System (BMS) is responsible for monitoring individual cell voltages and temperatures in real-time, as a safety requirement and as a necessary input for the State of Charge and State of Health estimation. However, achieving an accurate real-time state estimation remains a challenge, particularly under the dynamic current/voltage profiles of a race.

This work addresses this challenge by developing and validating a model-based dual estimation framework for simultaneous SOC and SOH estimation of a 30 Ah / 111 Wh NMC battery [3] used in the eRacing electric motorcycle. A second-order RC equivalent circuit model is parameterised through experimental characterisation, and dual Extended Kalman Filters are implemented to estimate battery states under dynamic operating conditions.

Introduction

High-performance electromobility relies heavily on Lithium-ion (Li-ion) battery technology due to its superior energy-to-weight ratio, low self-discharge rates, and high efficiency compared to other rechargeable technologies [4]. Nickel-Manganese-Cobalt (NMC) is a type of Li-ion battery. NMC cells are widely adopted for high-performance electric mobility applications due to their high energy density [8], which makes them suitable for motorsport applications since minimising the weight while maintaining high power capability is an essential design requirement. However, Lithium-ion cells require a robust system to ensure efficient operation within the Safe Operating Area; this embedded system is known as the Battery Management System (BMS).

The BMS is a critical component in any electric vehicle employing Li-ion batteries; its primary functions include detecting unsafe operating states, protecting cells from unsafe conditions, extending battery longevity, and ensuring that the battery pack operates within its design specifications [17]. To achieve these objectives, the system continuously measures, balances, reports, records, and estimates a set of vital parameters.

The importance of the BMS is reflected in international standards such as ISO 26262, which defines requirements for vehicle functional safety [11]. Failure to implement an effective BMS can lead to various hazards, including inaccurate SOC and SOH estimation, which may result in overcharging, thermal runaway, and accelerated cell degradation [11]. Consequently, cell voltages, battery-pack current, and external cell temperatures are continuously monitored and protection mechanisms are implemented when abusive operating conditions are detected. These measurements are essential for detecting faults and deploying protocols against overcharge/discharge, overcurrent, short circuits, and extreme temperatures [17]. Furthermore, these signals are input variables needed for state estimation algorithms, which are fundamental for optimising battery performance and predicting reliability.

To optimise the use of a Li-ion battery, an accurate assessment of the available energy and power at any given moment is essential, as these metrics determine the achievable driving range and performance of an electric vehicle. These parameters are closely related to the State of Charge (SOC) and the State of Health (SOH), which are two of the most important parameters managed by the BMS. However, a fundamental limitation is that these states cannot be directly measured by physical sensors, instead, they must be inferred [13]. Consequently, the BMS relies on estimation algorithms to calculate the SOC and SOH using external and observable signals, such as voltage, current, and temperature [7].

This limitation represents a significant challenge in battery management, as the accuracy of these estimations impacts the system efficiency, safety, and battery lifetime [24]. SOC is defined as the percentage of the remaining capacity of a battery cell relative to its nominal capacity [13]. Accurately estimating this value is difficult, as SOC is related to the concentration of lithium in the negative electrode [17] and direct measurement is not possible.

As a result, State of Charge must be inferred from observable variables, while also being influenced by multiple factors including hysteresis effects, temperature, aging, and charge/discharge rates, efficiency, and number of cycles [24]. Furthermore, dynamic operating conditions in a motorsport context such as high and rapidly varying current loads and high thermal variability further complicate the estimation process [17].

The State of Charge cannot be directly inferred from voltage measurements, as SOC depends on particle average concentrations, whereas voltage is related to surface concentrations, which are

influenced by temperature. In contrast, SOC changes only as a result of current flowing through the battery, either during the charge or discharge process [17].
Different approaches have been proposed in the literature to estimate the State of Charge. One of the simplest to implement is Coulomb Counting, in which SOC is computed by integrating the battery current over time. While this method is straightforward to implement and demands low computational effort, it presents cumulative errors due to sensor inaccuracies, current assumptions, and lack of reliable initial state value [17]. An alternative approach is through a lookup table based on the relationship between open-circuit voltage (OCV) and SOC, obtained by experimental characterisation under controlled conditions. Although this method can provide accurate results and is often integrated into more complex algorithms, it requires long relaxation periods to measure the true OCV, proving impractical for applications requiring real-time monitoring or involving dynamic loads [12].
More complex alternatives include Machine Learning and Deep Learning techniques. These methods do not require characterising the internal battery dynamics and can effectively handle the nonlinearities of the system. However, their performance relies heavily on the quality and availability of extensive training data [4]. Model-based approaches, by contrast, improve estimation accuracy by incorporating battery dynamics explicitly, accounting for sensor noise, and correcting estimates recursively using real-time measurements.
These methods do not require large training datasets, making them suitable for applications where experimental data is limited. Among these methods, Kalman filter-based algorithms, such as Kalman Filter (KF), Extended Kalman Filter (EKF), or Unscented Kalman Filter (UKF), demonstrate competitive performance in the literature. These algorithms use a mathematical model of the battery to propagate state estimates, while simultaneously computing the uncertainty associated with those predictions through a covariance matrix [7].
Unlike data-driven techniques, model-based estimation methods require an explicit mathematical representation of the battery. Consequently, the performance of Kalman filter algorithms depends strongly on the accuracy of the battery model. The two main modelling approaches are physics-based electrochemical models and equivalent circuit models (ECMs).
Physics-based models describe the electrochemical processes occurring within the cell and can provide highly accurate representations of battery behaviour. However, they require detailed knowledge of the cell chemistry and internal structure, resulting in high computational complexity and implementation effort. In contrast, equivalent circuit models represent the battery using electrical components such as voltage sources, resistors, and capacitors. Although these models provide a simplified description of the battery dynamics, they offer a suitable trade-off between accuracy and computational efficiency, making them particularly suitable for online battery management applications.
Accurate SOH estimation constitutes a significant challenge in battery management systems and is a fundamental indicator of cell degradation, reflecting the declining performance capabilities of the battery pack in its lifetime [17]. The two metrics most widely associated with battery health are the total capacity of the cell and the equivalent series resistance (ESR).
Variations in these quantities are caused by unwanted chemical reactions and structural degradation. Specifically, anode aging predominantly drives the increase in the ESR, whereas cathode degradation is associated with total capacity reduction [17]. Consequently, accurate SOH estimation is essential for evaluating battery lifetime, reliability, and operational safety. A cell is commonly considered to have reached its end-of-life when its capacity reaches 80% of its nominal value, and is therefore replaced to ensure safe and reliable operation [23].
Similar to SOC, the SOH cannot be directly measured and therefore must be inferred from observable signals, such as current, voltage, temperature, and SOC estimates. Nonlinear Kalman filters and machine learning techniques are commonly used for SOH estimation. However, in the literature, SOH estimation methods are frequently integrated with SOC estimation frameworks through joint or dual estimation approaches combined with Extended Kalman or Sigma-Point Kalman filters. The joint estimation approach employs an augmented state vector to simultaneously estimate SOC and SOH, whereas the dual estimation approach uses separate nonlinear filters operating in parallel [17].

This work proposes a model-based SOC and SOH estimation framework for the Nickel-Manganese-Cobalt cell used in the eRacing electric motorcycle at ETSEIB. A second-order RC equivalent circuit model is parameterised through experimental characterisation of the WeLion SHP270-30 cell [3]. A dual Estimation framework using the Extended Kalman Filter is implemented and validated using real charge-discharge data.

Chapter 1

State Of Art

The accurate estimation of battery internal states is an active research area driven by the growing adoption of electric vehicles, energy storage systems, and innovation in lithium-ion batteries. The literature can be divided intro three approaches: the empirical, model-based and data-driven methods, each approach differs in accuracy, computational cost, and implementation complexity [23]. While reviewing estimation approaches, it is necessary to also consider the battery models used to describe the battery's dynamics.

1.0.1 Empirical Estimation Methods

One of the simplest models for SOC estimation is the Coulomb Counting (CC), also known as Ampere-hour integration method. This is an open-loop approach that computes the State of Charge by integrating the measured cell current over time. The advantages of this method rely on its simple implementation and low computational demand [7], nonetheless, its performance is highly dependent on the initial SOC characterisation, furthermore CC is susceptible to sensor inaccuracies and accumulated integration errors [13][12].

A complementary approach to CC is the Open-Circuit Voltage (OCV) method, which applies the relationship between OCV and SOC to infer the battery state from a voltage measurement. This OCV-SOC curve is commonly characterised experimentally and stored as a lookup table [23]. Despite its simplicity and applicability across different battery chemistries, the OCV-SOC relationship only holds accurately when the cell is fully relaxed and undisturbed for several hours p rior to measurement. This condition can only be achieved under controlled laboratory conditions, making the method unsuitable for applications involving dynamic loads [17]. As a result, both the CC and OCV are typically integrated into more complex estimation methods than used independently.

1.0.2 Battery Modelling Approaches

To review model-based estimation techniques, different types of battery models need to be considered first, as the choice of model conditions the achievable accuracy and the feasibility of the implementation.

In the literature there are two different methods to model a battery, the Physics-based models (PBM) and the Equivalent Circuit models (ECM) [17]. PBMs capture the internal dynamics of a battery cell by constructing the equations from electrochemical and thermodynamic laws, which describe the physical transport and diffusion and electrochemical processes occurring in the cell. This type of model offers high predictive accuracy across a wide range of operating conditions and insights into the internal electrochemical states, useful for predicting ageing. However, the mathematical complexity involves coupled partial differential equations and demands high computational effort, which results in high costs and multiple challenges to implement on an embedded BMS [17].

ECMs represent the battery cell as an electric circuit composed of resistors, capacitors, and an ideal voltage source, providing an approximate behaviour of the current-voltage dynamics. Its accuracy is generally lower than that of physics-based models, and its predictive validity is restricted to the specific operational conditions in which it was characterised [17] [23]. However, its simple structure, parameterisation process, and suitability to implement in real applications offer significant practical advantages. Although the parameterisation process demands systematic and methodical laboratory testing, the model's low mathematical complexity makes it more feasible to implement in comparison to PBMs. The parameter values of an ECM depend on operating conditions such as temperature, C-rate, depth of discharge, and ageing status, which must be accounted for in the model structure [5]. Among ECM variants, the second-order Thévenin model has emerged as the standard choice in the literature. This circuit comprises of an OCV source in series with a resistor connected in series with two parallel branches, each with a capacitor and a resistor. This model captures both electrochemical polarisation and concentration polarisation dynamics while remaining computationally advantageous for BMS applications [7][13][12].

1.0.3 Adaptive and Fuzzy Logic Methods for Estimation

To improve on fixed-parameter limitations of the standard model-based estimators, works have proposed adaptive strategies that update model parameters. The main challenge is the online parameter identification, traditional offline methods based on least-squares curve fitting introduce accuracy limitations, while standard online approaches such as the Recursive Least Squares (RLS) algorithm suffer from data saturation and sensitivity to initial value settings, making it difficult to reliably track time-varying parameters [20]. Zhang et al. [24] demonstrated that by incorporating dynamic, SOC-dependent parameters identified through limited-memory recursive least-squares algorithm, the results show a more representative model of the actual battery discharge behaviour, with the resulting UKF-based SOC estimates closely matching those obtained via the ampere-hour method used as a reference benchmark.

A further extension of model-based estimation involves the integration of fuzzy logic to dynamically adjust filter parameters during operation. Wang et al. [20] proposed combining the Multiple Innovation Least Squares (MILS) algorithm for online parameter identification with a fuzzy-logic mechanism that continuously adapts the observation noise covariance matrix of an EKF. The MILS algorithm proves to enhance the data efficiency and convergence speed while the fuzzy-logic optimises the balance between the time update and measurement correction of the filter to improve higher precision under the coupled and dynamic conditions of an electric vehicle. This type of hybrid methods represents a more complex implementation, but addresses some of the robustness limitations of the Kalman filter approaches.

1.0.4 Data-Driven and Machine-Learning Models

Growing interest in data-driven methods has risen as an alternative to model-based approaches, due to their effectiveness in handling nonlinearities without requiring characterising the internal battery dynamics. These methods require experimental data when training to learn the input/output relationship [23]. Neural-Network estimators, such as the backpropagation (BP) have been applied to SOC across different battery chemistries and operating conditions [22]. Xu et al. demonstrated that for low-speed electric vehicle batteries, BP neural networks that have sufficiently deep network topologies can achieve suitable SOC prediction accuracy during charging cycles, with the understanding that results are sensitive to the network architecture and the size and diversity of the training dataset [22].

In Bosello's work [4] different deep learning approaches are provided for Li-ion battery state estimation, demonstrating the competitive accuracy across a range of conditions. Nonetheless, the consistent limitation in all data-driven methods is their dependency on large and high quality training datasets that must represent the operating conditions faced [13] [4]. Consequently, for

applications where training data for the specific cell under its operating conditions is insufficient, model-based approaches are more practical.

1.0.5 Kalman Filter Methods for SOC and SOH Estimation

Kalman filter algorithms have demonstrated consistency, feasibility, and cost-effective performance as battery estimators [13]. The Linear Kalman Filter is optimal for linear systems with Gaussian noise but cannot directly accommodate the nonlinearities of a lithium-ion OCV/SOC relationship [12]. The Extended Kalman Filter addresses this by linearising the nonlinear model at each time step through a first-order Taylor expansion, followed by the standard recursive Kalman filter [17].

Luo et al. implemented an EKF combined with a second-order RC ECM for Li-ion batteries on an electric vehicle, validating the algorithm under different current profiles: under constant current discharge with a maximum estimation error of 0.9%, under constant pulse and periodic variable current discharges where the error stayed within 1.2% and 1.8% respectively, and under rapidly varying current discharge, obtaining an error of 3.4% [13]. Gjengedal et al. applied a comparable EKF approach to a Formula Student electric race car, using lithium-polymer cells which were subjected to high and rapidly varying current profiles. Their implementation achieved a SOC estimation error below 0.66%, proving that Kalman filter estimators are a suitable option for motorsport applications [7].

The Sigma-Point or Unscented Kalman Filter (UKF) avoids the approximation error introduced by the EKF's linearisation step by performing a statistical linearisation at each time step rather than computing Jacobians [17]. It is concluded that the UKF avoids the linearisation error inherent to the EKF and achieves higher estimation precision, at the cost of increased computational complexity.

For simultaneous estimation of the SOC and SOH parameters, the joint and dual estimation techniques have been developed. Liu et al. have proposed a joint online SOC and SOH estimation method using a Dual EKF, in which one filter estimates SOC at a fast sampling period of 1 second while a second filter estimates the SOH-related parameters at 60-second intervals [12]. The method was tested using a second-order RC model of a ternary lithium cell under both constant current and random current profiles, achieving SOC and SOH estimation accuracy that outperformed standalone EKF and UKF implementations.

1.0.6 Contribution

The reviewed literature demonstrates that an Extended Kalman Filter approach combined with a second-order RC equivalent circuit model provides a validated and feasible method to estimate the State of Charge of a battery under dynamic conditions. Incorporating UKF, adaptive parameter identification, and dual SOC/SOH estimation have further improved the accuracy and robustness of the estimation. However, these adjustments have largely been studied in isolation from one another and from the specific demands of motorsport applications. Gjengedal et al. [7] validated an EKF SOC estimator on a Lithium Polymer (Li-Po) cell under race current profiles, but did not extend the framework to SOH estimation. Similarly, Liu et al. [12] demonstrated a Dual EKF approach for joint SOC and SOH estimation on a ternary lithium cell for stationary energy storage and not for a motorsport application. Furthermore, conventional parameter identification methods used across the reviewed studies require extended rest periods between tests, a limitation that the algebraic method of Devarakonda et al. [5] is specifically designed to overcome.
This work combines the experimental characterisation of the WeLion SHP27030 cell, efficient parameter identification through the algebraic method, and a Dual EKF approach for joint SOC and SOH estimation, intended for an eRacing electric motorcycle running under dynamic operating conditions.

Chapter 2

Methodology

2.1 Battery Model

The Battery Management System (BMS) is responsible for calculating estimates of critical internal variables, such as the State of Charge (SOC) and State of Health (SOH). Consequently, implementing a high-fidelity mathematical framework that captures the complex dynamics of the battery is essential. A widely adopted approach is the equivalent-circuit model (ECM), in which the behaviour of a lithium-ion cell is represented as an electrical circuit. Such model requires the characterisation of specific parameters and cell dynamics through conducted tests under controlled conditions [17].

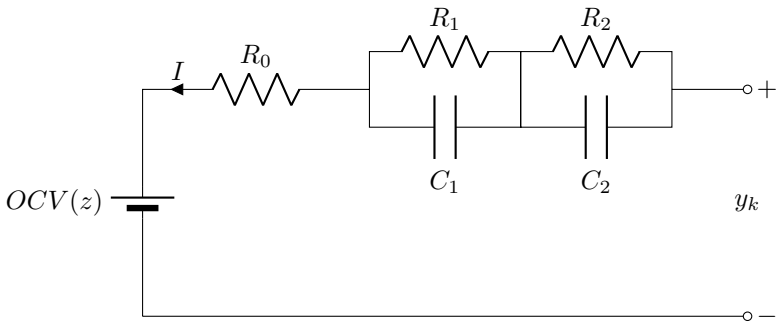


Figure 2.1: Second-order Thévenin equivalent circuit model.

The discrete-time ECM used in this study is defined by three primary components: the State of Charge, the diffusion currents within the RC branches, and the hysteresis voltage.
The State of Charge at a discrete-time index k, denoted as z_k, represents the ratio of the remaining capacity to the nominal capacity of the cell. Its temporal evolution is governed by the following relationship:

$$z_{k+1} = z_k - \eta \frac{\Delta t}{Q}(i_k + w_k) \tag{2.1}$$

For this study, the battery is represented as a second-order Thévenin equivalent circuit, as shown in Figure 2.1. The polarisation and diffusion effects are modelled using two resistor-capacitor networks. The currents circulating through the resistors R_1 and R_2 at time step k are denoted as $i_{R1,k}$ and $i_{R2,k}$ respectively, and their discrete-time evolution is described by:

$$i_{R1_{k+1}} = \tau_1 i_{R1_k} + (1 - \tau_1)(i_k + w_k) \tag{2.2}$$

$$i_{R2_{k+1}} = \tau_2 i_{R2_k} + (1 - \tau_2)(i_k + w_k) \tag{2.3}$$

Where the time constants τ_1 and τ_2 are defined as:

$$\tau_1 = e^{-\frac{\Delta t}{(R_1 C_1)}} \tag{2.4}$$

$$\tau_2 = e^{-\frac{\Delta t}{(R_2 C_2)}} \tag{2.5}$$

The third component is the hysteresis state h_k. The discrete-time evolution of the hysteresis state is modelled as:

$$h_{k+1} = e^{\left(-\left|\frac{\eta_k i_k \gamma \Delta t}{Q}\right|\right)} h_k + (e^{\left(-\left|\frac{\eta_k i_k \gamma \Delta t}{Q}\right|\right)} - 1)\text{sgn}(i_k) \tag{2.6}$$

In this work, the hysteresis state is excluded. Under the specific operating conditions considered for this application, hysteresis effects are small, and omitting this state reduces the computational complexity of the model without incurring a significant loss in estimation accuracy.
Finally, the terminal output voltage y_k is defined by the following observation equation:

$$y_k = OCV(z_k) - R_1 i_{R1_k} - R_2 i_{R2_k} - R_0 i_k + v_k \tag{2.7}$$

z_k is the state of charge for time k.
i_k is the current exchanged by the battery at its connection in [A].
$i_{R1_{k+1}}$ is the current through the resistance 1 for the time k in [A].
$i_{R2_{k+1}}$ is the current through the resistance 2 for the time k in [A].
y_k is the voltage across the terminals of the battery for the time k in [V].
η is the Coulombic efficiency.
Δt is the time step.
R_0 is the pure ohmic internal resistance of the battery in [Ω].
R_1 and R_2 are the resistance for the RC block 1 and 2 in [Ω].
C_1 and C_2 are the capacitance for the RC block 1 and 2 in [F].
$OCV(z_k)$ is the Open Circuit Voltage corresponding to the state of charge z_k in [V].
Q is the Electric capacity in [As].

2.1.1 Racing Motorcycle Battery Model

The experimental methodology of this work focuses on the battery used to power the electric racing motorcycle developed at ETSEIB. The specific cell model used is the WeLion SHP270-30 [3], a type of Nickel Manganese Cobalt (NMC) lithium-ion pouch cell with a nominal capacity of 30 Ah and a nominal energy of 111 Wh. NMC cells are typically used in high-performance racing applications due to their high energy density and favourable energy-to-weight ratio, as well as in electric vehicles [8]. This cell was selected for both the experimental characterisation and the implementation of the estimation algorithms.

2.2 OCV-SOC Curve Characterisation

The OCV-SOC curve describes the relationship between the Open-Circuit Voltage (OCV) and the SOC of the cell. This relationship is a key element of the battery model described in Section 2.1, since it is necessary for the output equation (2.7) and to compute the differential term $dOCV/dz$ used in the Extended Kalman Filter (EKF) implementation shown in Section 2.4.
Three separate experimental tests are required to obtain the OCV-SOC curve. Through the first two tests, described in Sections 2.2.1 and 2.2.2, the cell capacity, exchanged energy, and efficiency are computed. For all three tests, the terminal voltage, current and time must be continuously recorded using a power analyser. The tests are as follows:

2.2.1 Discharge Test

The discharge test must be conducted at a low constant current to minimise internal resistance losses, ensuring the capacity measurement accurately reflects the true electrochemical capacity of the cell. Due to the safety risks inherent to handling Li-ion batteries, the test requires continuous laboratory supervision. Consequently, the magnitude of the discharge current is constrained by the maximum safe operating window and the nominal capacity of the cell under test. This operational boundary is established as follows:

$$I_{dis} \leq \frac{Q_{nom}}{t_{dis}} \tag{2.8}$$

where Q_{nom} represents the nominal capacity of the cell and t_{dis} is the maximum permissible duration for continuous supervision. Based on this constraint, a conservative constant discharge current I_{dis} is selected to guarantee the test concludes within the monitored timeframe.
From the recorded current profile and the total discharge duration t_{dis}, the actual discharge capacity of the cell Q is determined by,

$$Q = I_{dis} t_{dis} \tag{2.9}$$

The total energy released by the cell during the discharge process is theoretically defined by integrating the instantaneous power over the total test duration as,

$$E_{dis} = \int_0^{t_{dis}} v(t)\, i(t)\, dt \tag{2.10}$$

In practice, because data collection relies on discrete time sampling, the total energy is calculated numerically. Assuming a constant sampling interval Δt, the discrete energy calculation is expressed as:

$$E_{dis} = \sum_{n=1}^{N} v_n i_n \Delta t \tag{2.11}$$

Where the parameters from equations 2.9, 2.10, and 3.3 are defined as follows,

Q Actual discharge capacity of the cell [Ah].

I_{dis} Constant discharge current magnitude [A].

t_{dis} Total duration of the discharge test [h].

E_{dis} Total energy released during discharge [Wh].

$v(t), v_n$ Continuous and discrete measured terminal voltage [V].

$i(t), i_n$ Continuous and discrete measured discharge current [A].

Δt Sampling time step expressed in hours.

N Total number of recorded data samples.

2.2.2 Charge Test

In contrast to the discharge test, the charging procedure does not maintain a single constant current throughout its duration. Instead, it follows a standard Constant-current Constant-voltage (CC-CV) protocol, which is critical for safely maximising the SOC without exceeding the cell's upper voltage limit. This behaviour is illustrated in Figure 2.2.

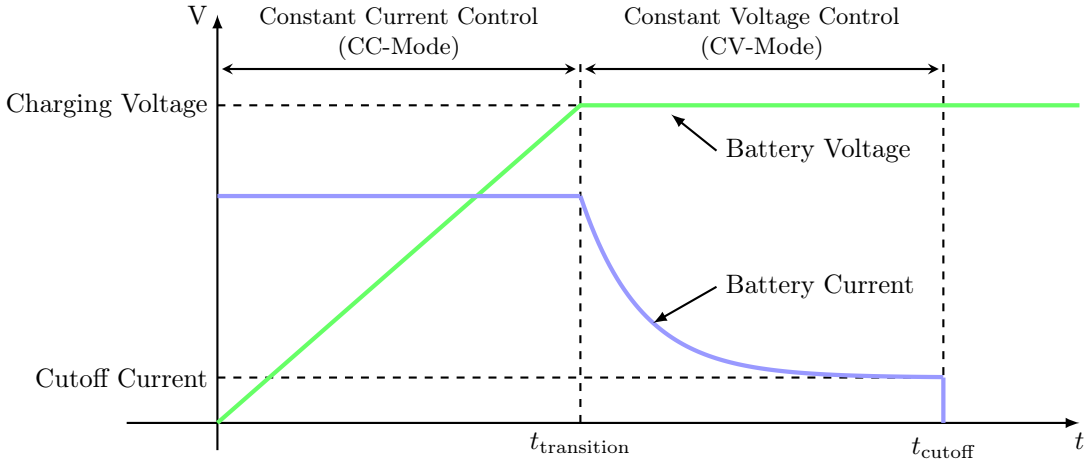


Figure 2.2: Generic CC-CV charging profile characterisation curve [10].[1]

During the constant current mode, a steady current pushes energy into the cell, causing the terminal voltage to rise linearly. Once the cell reaches its upper charging threshold, the control system switches to constant voltage mode; the voltage is clamped, and the current exponentially decays. The total energy consumed by the cell during this charging sequence (E_{charge}) is computed by integrating the continuous power profile:

$$E_{charge} = \int_0^{t_{charge}} v(t)\, i(t)\, dt \tag{2.12}$$

Mirroring the discrete formulation used in the discharge analysis, the practical computation from experimental dataset values is computed as:

$$E_{charge} = \sum_{n=1}^{M} v_n i_n \Delta t \tag{2.13}$$

where M represents the total number of sampling points accumulated across both charging phases. With both experimental energy parameters quantified, the net energy efficiency (η) of the battery system is calculated as,

$$\eta = \frac{E_{dis}}{E_{charge}} \tag{2.14}$$

2.2.3 Pulses Test

Characterising the OCV vs. SOC relationship requires executing an intermittent pulse discharge test. This profile applies consecutive, discrete discharge pulses designed to deplete the cell capacity by a precise, predetermined percentage step.

[1] During the preparation of this work, the author used Gemini 3.5 to produce Figure 2.2. After using this tool, the author reviewed and edited the content as necessary, and she assumes full responsibility for the content of the publication.

Each active current pulse is followed by an extended relaxation period, allowing the internal chemical diffusion transients to decay and the terminal voltage to stabilise to its true equilibrium OCV state. Using the actual capacity Q obtained from the continuous discharge test (Section 2.2.1), the capacity step required for each specific depth of discharge interval, Q_{step} is expressed by:

$$Q_{step} = Qp_{step} \tag{2.15}$$

$$t_{dis_{5\%}} = \frac{Q_{5\%}}{I_{cons}} 60 \tag{2.16}$$

where p_{step} represents the chosen fractional capacity drop per pulse. To implement this configuration in the cycler software under a chosen constant test current (I_{cons}), the required active pulse duration (t_{pulse}) in minutes is calculated as:

$$t_{pulse} = \frac{Q_{step}}{I_{cons}} 60 \tag{2.17}$$

where t_{pulse} represents the explicit window length in minutes during which the current I_{cons} must be drawn to accurately shift the cell down the SOC tracking curve by the designated interval.
The sequential current profile configuration utilised to execute this experimental pulse sequence is illustrated schematically in Figure 2.3.

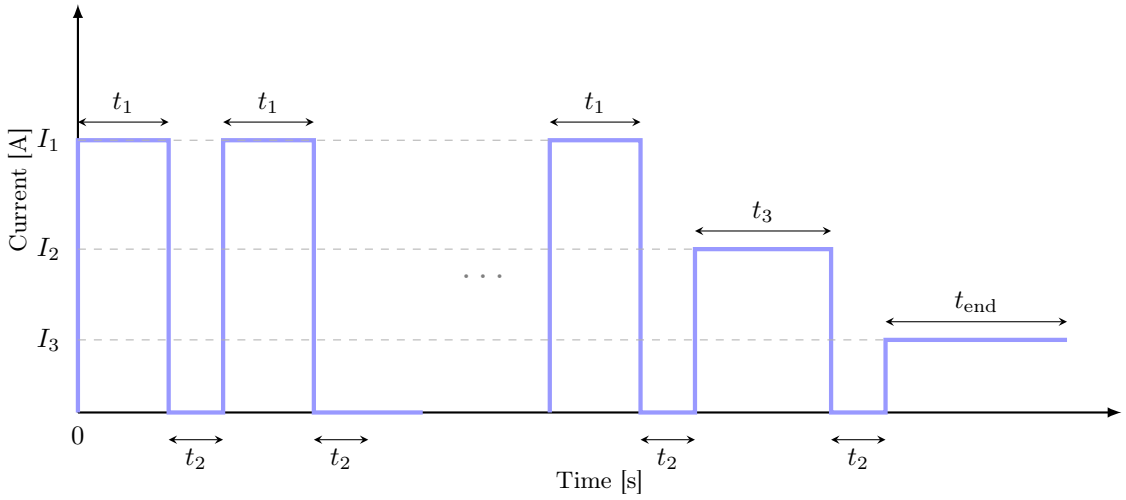


Figure 2.3: Representation of the current pulses.[2]

As shown in the profile, the core characterisation routine consists of discrete active discharge periods of magnitude I_1 applied for a strict calculated duration t_1. Each individual current interaction is followed by an invariant relaxation period t_2 at zero current input 0 A.
To accommodate changing operational bounds as the battery nears empty SOC states, the profile incorporates an intermediate parameter stage featuring an altered current step magnitude I_2 maintained across an altered duration window t_3.
Finally, the total diagnostic routine transitions into a continuous constant-current depleted phase of magnitude I_3 for a remaining duration t_{end}, ensuring the battery system safely reaches its terminal lower voltage cut-off boundary.

[2]During the preparation of this work, the author used Gemini 3.5 to produce Figure 2.3. After using this tool, the author reviewed and edited the content as necessary, and she assumes full responsibility for the content of the publication.

2.3 Parameter Identification

The accuracy of the ECM is critically dependent on the precise identification of its internal parameters. As described in Section 2.1, a second-order Thévenin equivalent circuit was selected for this work, therefore, the parameters to be identified are the Open Circuit Voltage, the Ohmic resistance R_0, and the resistances and capacitances of the two RC branches, R_1, C_1, R_2, and C_2. The conventional approach to parameter identification is the least-squares curve fitting of experimental voltage responses obtained during controlled charge and discharge cycles. A fundamental assumption of this method is that each test starts from zero initial conditions, which requires a rest period of 30 minutes between tests [5], making the full characterisation of a battery a time-consuming and impractical process.
To overcome this limitation, the algebraic parameter identification method proposed by Devarakonda and Hu [5] is followed in this work. The key contribution of this method is that it extracts all circuit parameters simultaneously with the unknown initial conditions of the capacitor voltages, thereby eliminating the requirement for relaxation periods between tests and accelerating the characterisation process.
The process can be divided into four stages, which will be described in Section 2.3.2.

2.3.1 Parameter Identification - Test Setup

The setup for the parameter identification requires an electronic load (as in Figure 2.4) to draw a step discharge current I which will change to I_1 (as described in (2.18)) and the terminal voltage $v(t)$ to be recorded continuously.

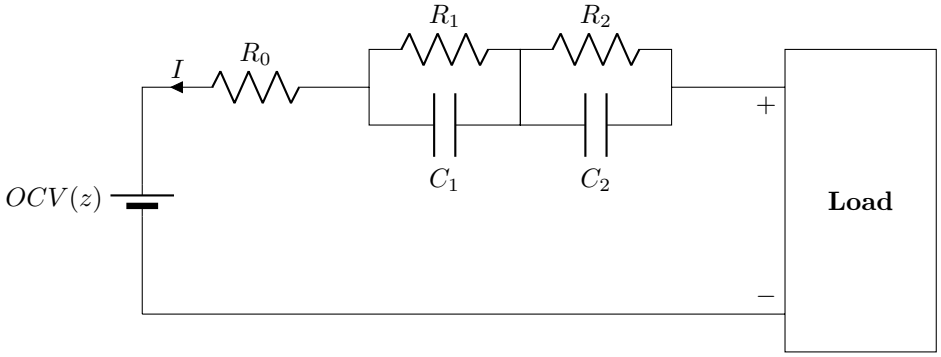


Figure 2.4: Second-order Thévenin equivalent circuit model connected to a load.

$$i(t) = \begin{cases} I, & 0 < t \leq T_0 \\ I_1, & t > T_0 \end{cases} \tag{2.18}$$

The full parameter set is recovered from exactly eight voltage samples taken at the following instants:

$$v(0^+),\ v(T),\ v(2T),\ v(3T),\ v(4T),\ v(T_0^+),\ v(T_0 + T_1),\ v(T_0 + 2T_1) \tag{2.19}$$

where $T \in (0, T_0/4]$ is a sampling interval and $T_1 > 0$ is a second sampling interval used after the current step.

2.3.2 Test Steps

Step 1: Identification of R_0

The Ohmic resistance R_0 is responsible for the instantaneous voltage drop when a load is connected, since capacitors cannot change their voltage instantaneously. It is obtained as:

$$R_0 = \frac{v(0^-) - v(0^+)}{I} \tag{2.20}$$

where $v(0^-)$ is the terminal voltage immediately before the load is applied and $v(0^+)$ is the first sample after the current begins to flow.

Step 2: Obtaining time constants and initial condition combinations

Under the constant current I during the interval $(0, T_0)$, the terminal voltage evolves as:

$$v(t) = (E - R_0 I - R_1 I - R_2 I) + s_1\, e^{-t/\tau_1} + s_2\, e^{-t/\tau_2} \tag{2.21}$$

where $(E - R_0 I - R_1 I - R_2 I)$ is constant and $\tau_1 = R_1 C_1$, $\tau_2 = R_2 C_2$ are the time constants of the RC branches. The terms $s_1 = v_{10} - R_1 I$ and $s_2 = v_{20} - R_2 I$ combine the unknown initial capacitor voltages v_{10} and v_{20} with the corresponding resistances.
Taking differences between consecutive samples to eliminate the unknown constant term as:

$$b_k = v(k-1)T - v(kT), \qquad k = 1, 2, 3, 4 \tag{2.22}$$

Each b_k is a linear combination of two geometric sequences with common ratios $d_1 = e^{-T/\tau_1}$ and $d_2 = e^{-T/\tau_2}$, which satisfies the second-order linear recurrence:

$$b_{k+2} = (d_1 + d_2)\, b_{k+1} - (d_1 d_2)\, b_k \tag{2.23}$$

Defining $u_1 = d_1 + d_2$ and $u_2 = d_1 d_2$, equation (2.23) applied to the measured differences yields the following linear system:

$$\begin{bmatrix} b_2 & -b_1 \\ b_3 & -b_2 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} b_3 \\ b_4 \end{bmatrix} \tag{2.24}$$

The decay factors d_1 and d_2 are then recovered as the two real roots of the characteristic polynomial:

$$d^2 - u_1\, d + u_2 = 0 \tag{2.25}$$

from which the time constants follow as:

$$\tau_1 = -\frac{T}{\ln d_1}, \qquad \tau_2 = -\frac{T}{\ln d_2} \tag{2.26}$$

With d_1 and d_2 known, the amplitudes s_1 and s_2 are obtained by solving:

$$\begin{bmatrix} 1 & 1 \\ d_1 & d_2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}, \qquad s_j = \frac{x_j}{d_j - 1}, \quad j = 1, 2 \tag{2.27}$$

Step 3: Obtaining R_1, R_2, C_1, C_2 from the current step

At time T_0, the capacitor voltages have evolved to:

$$v_j(T_0) = s_j\, a_j + R_j I, \qquad a_j = e^{-T_0/\tau_j}, \quad j = 1, 2 \tag{2.28}$$

After the current switches to I_1, the voltage response for $t > T_0$ is again a sum of two decaying exponentials. Applying the same differencing technique to the three post-step samples, and defining $p_j = e^{-T_1/\tau_j}$, the following system is obtained:

$$\begin{bmatrix} p_1 - 1 & p_2 - 1 \\ p_1(p_1 - 1) & p_2(p_2 - 1) \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} v(T_0^+) - v(T_0 + T_1) \\ v(T_0 + T_1) - v(T_0 + 2T_1) \end{bmatrix} \tag{2.29}$$

Since a_j, p_j, and s_j are all known from previous steps, the resistances are obtained as:

$$R_j = \frac{y_j - s_j\, a_j}{I - I_1}, \qquad j = 1, 2 \tag{2.30}$$

The remaining parameters then follow:

$$C_j = \frac{\tau_j}{R_j}, \qquad v_{j0} = s_j + R_j I, \qquad E = v(0^-) + v_{10} + v_{20} \tag{2.31}$$

Step 4: Optimisation of the sampling interval

In practice, experimental voltage measurements are subject to noise and the battery behaviour deviates from the ideal circuit model. As a result, the extracted parameters depend on the choice of T and T_1. To obtain the most accurate parameter set, a one-dimensional sweep over admissible values of T is performed, with $T_1 = T$. For each candidate value, the model voltage response is reconstructed from the extracted parameters and compared to the experimental response. The value of T that minimises the Root Mean Square Error (RMSE) between the two is selected:

$$T^* = \arg\min_{T} \mathrm{RMSE}(v_{\mathrm{exp}},\, v_{\mathrm{model}}) \tag{2.32}$$

2.4 Kalman Filter

Kalman Filter (KF) is a recursive model-based estimator that consists of a mathematical algorithm used to infer an unobservable internal state of a dynamic system from noisy measured variables [21]. As explained by Plett, a key parameter of the model is the Kalman filter gain L_k, which is time-variant and updated at each time step by continuously calculating the error covariance matrices to minimise the mean squared error [17].
To implement the KF, the system must be represented in a state-space form. For a linear system (or a system that can be assumed as linear), the state-space equations are defined as:

$$x_k = A_{k-1}x_{k-1} + B_{k-1}u_{k-1} + w_{k-1}, \tag{2.33}$$

$$y_k = C_k x_k + D_k u_k + v_k \tag{2.34}$$

where,
x_k is the state vector of the process in time step k.
y_k is the output.
u_k is the input at the time step k.
w_k and v_k are the process and measurement noise at time step k.

The KF algorithm is divided into two stages: time update (prediction) and measurement update equations (correction), as shown in Figure 2.5. The prediction equations project forward in time the current state and the error covariance estimates to obtain the a priori estimates for the next step. In the correction section, the algorithm incorporates the most recent measured values to compute the error between the observed and estimated output, producing an improved a posteriori state estimate [21].

The Kalman Filter algorithm assumes a linear system or a nearly linear system. However, because the relationship between OCV and SOC is nonlinear, applying a linear filter introduces significant

[3]During the preparation of this work, the author used Gemini 3.5 to produce Figure 2.5. After using this tool, the author reviewed and edited the content as necessary, and she assumes full responsibility for the content of the publication.

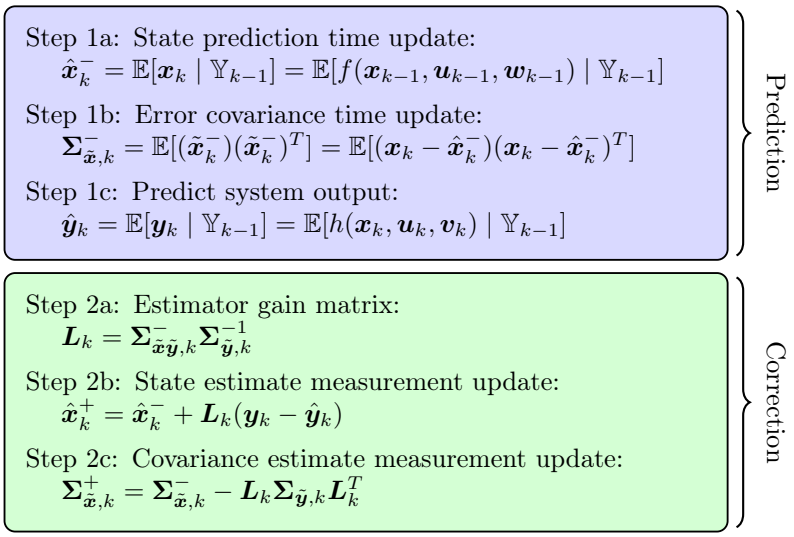


Figure 2.5: Prediction and Correction steps in Kalman Filter algorithm [17].[3]

estimation inaccuracies. Consequently, different methodologies capable of addressing these nonlinear dynamics must be employed to ensure a more robust estimation.
The nonlinear behaviour of the battery is shown in the following equations:

$$x_k = f(x_{k-1}, u_{k-1}, w_{k-1}), \tag{2.35}$$

$$y_k = h(x_k, u_k, v_k) \tag{2.36}$$

2.4.1 Extended Kalman Filter

An extension of the Linear Kalman Filter is developed to address these nonlinearities, known as the Extended Kalman Filter (EKF). This method linearises the nonlinear system at each time step while following the same prediction and correction steps as the Kalman Filter (as shown in Figure 2.6). The linearisation is achieved through the computation of the Jacobian matrices, the partial derivatives of the nonlinear functions with respect to the relevant quantities being estimated. For the SOC filter, these are the partial derivatives of the state and output equations with respect to the state vector $\hat{x}$, these matrices are shown as $\hat{A}_k$, $\hat{B}_k$, $\hat{C}_k$ and $\hat{D}_k$ in Section 2.6.1. For the SOH filter, the Jacobians are taken with respect to the parameter vector θ, and defined as $\hat{A}_k^\theta$, $\hat{B}_k^\theta$, $\hat{C}_k^\theta$, and $\hat{D}_k^\theta$ in Section 2.6.2. By evaluating these matrices at the state estimate, the EKF provides a linear approximation of the system's behaviour around the operating point [17].

2.5 Dual Estimation Technique

In this work, the dual estimation approach is employed to perform the state and parameter estimation simultaneously. Two separate EKF algorithms are implemented in parallel, each operating on its respective estimation problem while sharing state information at each time step. The SOC

[4]During the preparation of this work, the author used Gemini 3.5 to produce Figure 2.6. After using this tool, the author reviewed and edited the content as necessary, and she assumes full responsibility for the content of the publication.

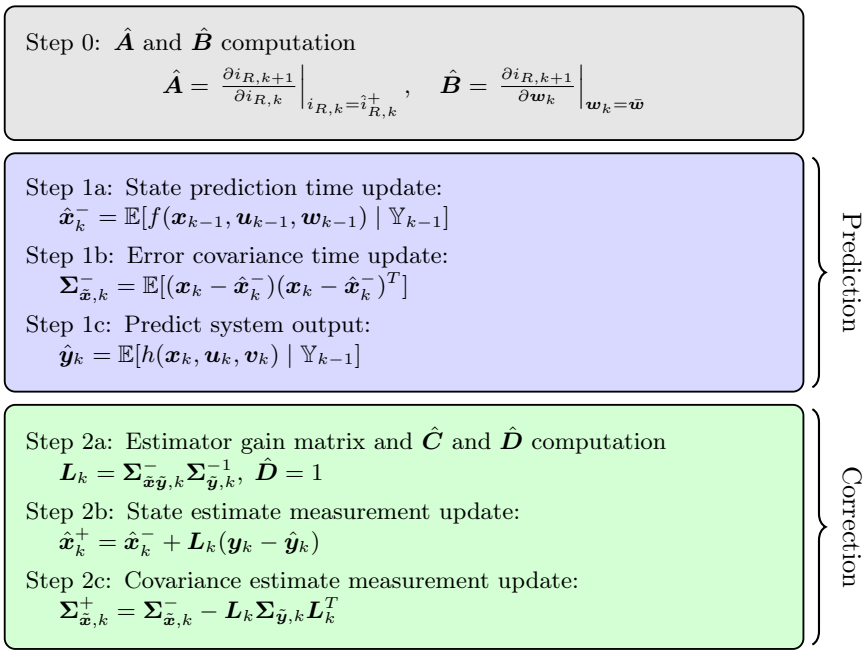


Figure 2.6: Prediction and Correction steps in an Extended Kalman Filter algorithm [17].[4]

estimator uses the updated ESR and capacity values provided by the SOH filter, while the SOH estimator incorporates the updated SOC estimate. The interaction between the filters is shown in Figure 2.7.

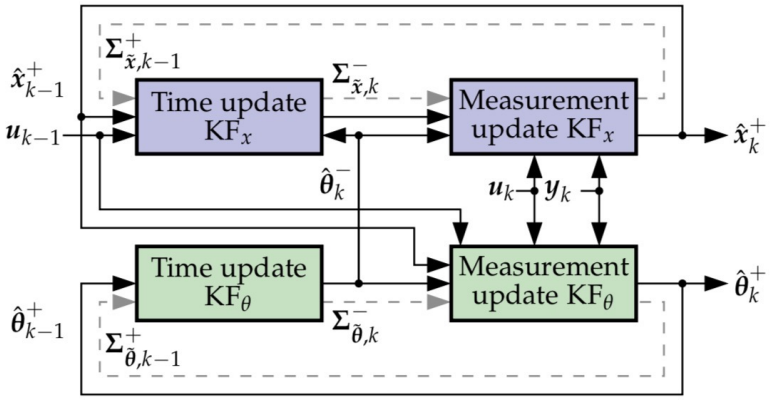


Figure 2.7: Interaction between SOC and SOH Extended Kalman Filters using Dual Estimation [17].

2.6 EKF Implementation

The implementation of the algorithms described in Sections 2.6.1, 2.6.2, and 2.6.3 and the steps shown in Section 2.6.4 are translated into MATLAB code[5], where the battery parameters and the

[5]During the preparation of this work, the author used Claude Sonnet 4.6 to produce and improve sections of the MATLAB code. After using this tool, the author reviewed and edited the content as necessary, and she assumes full responsibility for the content of the publication.

OCV/SOC curve values obtained following the methodology described in Sections 2.3 and 2.2, respectively, are stored.

2.6.1 EKF for SOC Estimation

To implement the EKF algorithm on a battery, the matrices $\hat{A}_k$, $\hat{B}_k$, $\hat{C}_k$ and $\hat{D}_k$ have to be determined based on the ECM cell model. These matrices represent the linearisation of the state equations (2.1), (2.2), and (2.3) previously defined in Section 2.1.
The state vector $\hat{x}$ comprises the currents flowing through the RC branches and the state of charge, which are the necessary variables to predict the terminal voltage according to equation (2.7):

$$\hat{x} = \begin{bmatrix} i_{R1} \\ i_{R2} \\ z \end{bmatrix} \tag{2.37}$$

Assuming a Coulombic Efficiency of $\eta = 1$, the SOC dynamics in equation (2.1) can be expressed as:

$$z_{k+1} = z_k - \frac{\Delta t}{Q}(i_k + w_k) \tag{2.38}$$

To linearise the model, the partial derivatives of the state equations with respect to the state vector and the process noise have to be calculated. From equation (2.38), the SOC-related derivatives are:

$$\left.\frac{\partial z_{k+1}}{\partial z_k}\right|_{z_k=\hat{z}_k^+} = 1 \tag{2.39}$$

$$\left.\frac{\partial z_{k+1}}{\partial w_k}\right|_{w_k=\bar{w}} = -\frac{\Delta t}{Q} \tag{2.40}$$

Similarly, the derivatives for the RC branch currents i_{R1} and i_{R2} are derived from equations (2.2) and (2.3), and are the following:

$$\left.\frac{\partial i_{R1,k+1}}{\partial i_{R1,k}}\right|_{i_{R1,k}=\hat{i}_{R1,k}^+} = \tau_1 \tag{2.41}$$

$$\left.\frac{\partial i_{R2,k+1}}{\partial i_{R2,k}}\right|_{i_{R2,k}=\hat{i}_{R2,k}^+} = \tau_2 \tag{2.42}$$

$$\left.\frac{\partial i_{R1,k+1}}{\partial w_k}\right|_{w_k=\bar{w}} = 1 - \tau_1 \tag{2.43}$$

$$\left.\frac{\partial i_{R2,k+1}}{\partial w_k}\right|_{w_k=\bar{w}} = 1 - \tau_2 \tag{2.44}$$

From previous equations (2.39), (2.41), and (2.42) matrix $\hat{A}$ is constructed as follows:

$$\hat{A}_k = \begin{bmatrix} \tau_1 & 0 & 0 \\ 0 & \tau_2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{2.45}$$

Similarly, matrix $\hat{B}$ is assembled from equations (2.40), (2.43), and (2.44), resulting in:

$$\hat{B}_k = \begin{bmatrix} 1-\tau_1 \\ 1-\tau_2 \\ -\frac{\Delta t}{Q} \end{bmatrix} \tag{2.46}$$

The observation matrices $\hat{C}_k$ and $\hat{D}_k$ are obtained by differentiating the measurement equation (2.7) with respect to the state vector and measurement noise:

$$\frac{\partial y_k}{\partial i_{R1}} = -R_1 \tag{2.47}$$

$$\frac{\partial y_k}{\partial i_{R2}} = -R_2 \tag{2.48}$$

$$\left.\frac{\partial y_k}{\partial z_k}\right|_{z_k=\hat{z}_k^-} = \left.\frac{\partial OCV(z_k)}{\partial z_k}\right|_{z_k=\hat{z}_k^-} \tag{2.49}$$

$$\frac{\partial y_k}{\partial v_k} = 1 \tag{2.50}$$

Thus, the resulting Jacobians from equations (2.47), (2.48), and (2.49) form $\hat{C}$ as:

$$\hat{C}_k = \begin{bmatrix} -R_1 & -R_2 & \frac{dOCV(z_k)}{dz} \end{bmatrix} \tag{2.51}$$

And $\hat{D}$ from equation (2.50) result in:

$$\hat{D}_k = \begin{bmatrix} 1 \end{bmatrix} \tag{2.52}$$

2.6.2 EKF for SOH Estimation

For SOH estimation, the EKF is formulated using the parameter vector defined in equation 2.54, which contains the parameters selected for estimation, the equivalent series resistance R_0 and the cell capacity Q. The dynamics of the parameters are modelled assuming that variations in R_0 and Q occur slowly over time and exclusively due to perturbations, therefore a noise input is introduced. This white noise term (r_{k-1}) does not represent a physical quantity, rather, it is introduced to model the gradual variation of the cell parameters resulting from battery ageing. The dynamic is defined as follows:

$$\theta_k = \theta_{k-1} + r_{k-1} \tag{2.53}$$

where,

$$\theta_k = \begin{bmatrix} R_0 \\ Q \end{bmatrix} \tag{2.54}$$

The measurement function used for SOH estimation is defined in equation 2.55 which derives from the output equation 2.36, with the addition of the parameters in θ and the sensor noise and modelling error described in e_k.

$$d_k = g(x_k, u_k, \theta, e_k) \tag{2.55}$$

For the SOH filter, the state and input matrices ($\hat{A}_k^\theta$ and $\hat{B}_k^\theta$) are identity matrices due to the dynamics in equation 2.53. In contrast, the matrices $\hat{C}_k^\theta$ and $\hat{D}_k^\theta$ must be calculated and are defined as follows:

$$\left.\frac{dg(x_k, u_k, \theta, e_k)}{de_k}\right|_{e_k=\hat{e}_k^-} = \hat{D}_k^\theta \tag{2.56}$$

$$\left.\frac{dg(x_k, u_k, \theta, e_k)}{d\theta}\right|_{\theta=\hat{\theta}_k^-} = \hat{C}_k^\theta = \frac{\partial g}{\partial \theta} + (\frac{\partial g}{\partial x_k})(\frac{dx_k}{d\theta}) \tag{2.57}$$

where,

$$g = OCV(z) - R_1 i_{R1} - R_2 i_{R2} - R_0 i_k \tag{2.58}$$

Further developing the terms for $\hat{C}_k^\theta$, taking into account equation 2.58,

$$\frac{\partial g}{\partial R_0} = -i_k \tag{2.59}$$

$$\frac{\partial g}{\partial Q} = 0 \tag{2.60}$$

obtaining the first term of $\hat{C}_k^\theta$ as:

$$\frac{\partial g}{\partial d\theta} = \begin{bmatrix} -i_k \\ 0 \end{bmatrix} \tag{2.61}$$

For the second term in $\hat{C}_k^\theta$, the partial derivative of g with respect to x_k is computed as:

$$\frac{\partial g}{\partial i_{R1}} = -R_1 \tag{2.62}$$

$$\frac{\partial g}{\partial i_{R2}} = -R_2 \tag{2.63}$$

$$\frac{\partial g}{\partial z} = \frac{dOCV}{dz} \tag{2.64}$$

resulting in,

$$\frac{\partial g}{\partial x_k} = \begin{bmatrix} -R_1 \\ -R_2 \\ \frac{dOCV}{dz} \end{bmatrix} \tag{2.65}$$

The final term required to compute $\hat{C}_k^\theta$ is recursive and obtained as,

$$\frac{dx_k}{d\theta} = \frac{\partial f}{\partial \theta} + (\frac{\partial f}{\partial x_{k-1}})(\frac{dx_{k-1}}{d\theta}) \tag{2.66}$$

Taking the state equations 2.2, 2.3, and 2.1, the corresponding partial derivatives with respect to R_0 and Q can be calculated as:

$$\frac{\partial i_{R1_k}}{\partial R_0} = 0 \tag{2.67}$$

$$\frac{\partial i_{R2_k}}{\partial R_0} = 0 \tag{2.68}$$

$$\frac{\partial z_k}{\partial R_0} = 0 \tag{2.69}$$

$$\frac{\partial i_{R1_k}}{\partial \theta} = 0 \tag{2.70}$$

$$\frac{\partial i_{R2_k}}{\partial \theta} = 0 \tag{2.71}$$

$$\frac{\partial z_k}{\partial \theta} = \frac{\Delta t i_{k-1}}{Q^2} \tag{2.72}$$

obtaining that,

$$\frac{\partial f}{\partial \theta} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & \frac{\Delta t i_{k-1}}{Q^2} \end{bmatrix} \tag{2.73}$$

The final term 2.66 captures the state evolution as a function of the parameters across time. $\frac{\partial f}{\partial x_{k-1}}$ inside this equation is the Jacobian of the state transition function with respect to the state vector evaluated at the previous estimate ($\hat{x}^+_{k-1}$), which is contained in the matrix $\hat{A}_{k-1}$ already obtained in the SOC estimation algorithm in Section 2.6.1.
Therefore substituting equation 2.66 as,

$$\frac{dx_k}{d\theta} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & \frac{\Delta t i_{k-1}}{Q^2} \end{bmatrix} + \hat{A}_{k-1}(\frac{dx_{k-1}}{d\theta}) \tag{2.74}$$

This recursive term is initialised at $k = 0$, since no prior information regarding the parameter is available at the beginning of the estimation process [17].
therefore $\hat{C^\theta_k}$ is,

$$\hat{C^\theta_k} = \begin{bmatrix} -i_k \\ 0 \end{bmatrix} + \begin{bmatrix} -R_1 \\ -R_2 \\ \frac{dOCV}{dz} \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & \frac{\Delta t i_{k-1}}{Q^2} \end{bmatrix} \tag{2.75}$$

For $\hat{D}_k^{\theta}$ the matrix is obtained as follows:

$$\frac{\partial g}{\partial e_k} = 1 \tag{2.76}$$

Resulting in,

$$\hat{D}_k^{\theta} = 1 \tag{2.77}$$

2.6.3 EKF Dual Implementation for SOC and SOH Estimation

In the dual estimation framework, both filters run simultaneously at each time step k, with the output of one serving as input to the other. The SOC filter requires the current parameter values θ_k^- to propagate the state and predict the terminal voltage accurately, since both the capacity Q in the SOC dynamics and the internal resistance R_0 in the output equation are elements of θ. To couple these two filters the equations 2.35 and 2.36 were modified as follows:

$$x_k = f(x_{k-1}, u_{k-1}, \theta, w_{k-1}) \tag{2.78}$$

$$y_k = h(x_k, u_k, \theta, v_k) \tag{2.79}$$

Similarly, the SOH filter requires an accurate estimate of the current state $\hat{x}_k^-$ to evaluate the output function 2.58 and to propagate the recursive matrix. The information exchanged at each time step starts at the SOH filter providing θ_k^- to the SOC filter before the state propagation step. The SOC filter provides $\hat{x}_k^-$, $\hat{x}_k^+$ and the Kalman gain L to the SOH filter, which uses them to evaluate the function shown in equation 2.58 to compute $\hat{C}_k^{\theta}$, and to update its algorithm. The complete execution order of the coupled algorithm is detailed in the following Section 2.6.4.

2.6.4 Recursive Algorithm

Once the Jacobian matrices are defined, as previously shown in sections 2.6.1 and 2.6.2, they have to be integrated into their corresponding Kalman Filter recursive algorithm and updated at each time step. The EKF algorithm goes through a recursive three-stage process which consists of updating the $\hat{A}_{k-1}$ and $\hat{B}_{k-1}$ matrices, then the time update (prediction), and measurement update (correction).

- **Step 0**
  - **SOC** - Update $\hat{A}_{k-1}$ and $\hat{B}_{k-1}$ (matrices shown in (2.45) and (2.46))
- **Step 1a** - Parameter Prediction Time Update
  - **SOC** - Propagate the state estimate forward using the nonlinear system dynamics and the measured current i_k, and update current Q value stored in θ vector.

    $$\hat{x}_k^- = f(\hat{x}_{k-1}^+, u_{k-1}, w_{k-1}, \theta_k^-) \tag{2.80}$$

  - **SOH** - Update the recursive sensitivity matrix $\frac{dx_k}{d\theta}$ by applying the updated $\hat{A}_{k-1}$ and $\frac{\partial f}{\partial \theta}$ values computed in Step 0.
- **Step 1b** - Error Covariance Time Update

- **SOC** - Project the uncertainty forward, accounting for the process noise covariance Σ_w.

$$\Sigma_{x,k}^{-} = \hat{A}_{k-1}\Sigma_{x,k-1}^{+}\hat{A}_{k-1}^{T} + \hat{B}_{k-1}\Sigma_w\hat{B}_{k-1}^{T} \tag{2.81}$$

- **SOH** - Predict the parameter uncertainty forward, accounting for the process noise covariance $\Sigma_{r,k-1}$

$$\Sigma_{\theta,k}^{-} = \Sigma_{\theta,k-1}^{+} + \Sigma_{r,k-1} \tag{2.82}$$

- **Step 1c** - Output Prediction
  - **SOC** - Calculate the predicted terminal voltage.

$$\hat{y}_k = h(\hat{x}_k^{-}, i_k, \theta_k^{-}) \tag{2.83}$$

  - **SOH** - Predicted output is evaluated using the prior state estimate $\hat{x}_k^{-}$ from the SOC filter and the prior parameter estimate θ_k^{-}.

$$\hat{d}_k = g(\hat{x}_k^{-}, u_k, \theta_k^{-}) = OCV(z_k^{-}) - R_1 i_{R1,k}^{-} - R_2 i_{R2,k}^{-} - R_0 i_k \tag{2.84}$$

- **Step 2a** - Estimator Gain Matrix
  - **SOC** - Update $\hat{C}_{k-1}$ and $\hat{D}_{k-1}$ matrices shown in (2.51) and (2.52) respectively and calculate the optimal gain L_k by weighing the model uncertainty against the measurement noise covariance Σ_v.

$$\Sigma_{y,k} = \hat{C}_k\Sigma_{x,k}^{-}\hat{C}_k^{T} + \hat{D}_k\Sigma_v\hat{D}_k^{T} \tag{2.85}$$

$$L_k = \Sigma_{x,k}^{-}\hat{C}_k^{T}(\Sigma_{y,k})^{-1} \tag{2.86}$$

  - **SOH** - Update $\hat{C}_{k-1}^{\theta}$ and $\hat{D}_{k-1}^{\theta}$ by deriving using chain rule as shown in Section 2.6.2 and obtain covariance and Kalman filter gain.

$$\Sigma_{d,k} = \hat{C}_k^{\theta}\Sigma_{\theta,k}^{-}(\hat{C}_k^{\theta})^{T} + \hat{D}_k^{\theta}\Sigma_e(\hat{D}_k)^{T} \tag{2.87}$$

$$L_k^{\theta} = \Sigma_{\theta,k}^{-}(\hat{C}_k^{\theta})^{T}(\Sigma_{d,k})^{-1} \tag{2.88}$$

- **Step 2b** - Parameter Estimate Measurement Update
  - **SOC** - Correct the predicted state using the residual between the measured voltage v_k and the predicted voltage $\hat{y}_k$.

$$r = y_k - \hat{y}_k \tag{2.89}$$

$$\hat{x}_k^{+} = \hat{x}_k^{-} + L_k r \tag{2.90}$$

  - **SOH** - Compute a posteriori R_0 and Q parameters stored in vector θ by obtaining the output prediction error r.

$$r = d_k - \hat{d}_k \tag{2.91}$$

$$\hat{\theta}_k^{+} = \hat{\theta}_k^{-} + L_k^{\theta} r \tag{2.92}$$

- **Step 2c** - Error Covariance Measurement Update
  - **SOC** - Update the covariance matrix to reflect the reduction in uncertainty after incorporating the measurement.

$$\Sigma_{x,k}^{+} = \Sigma_{x,k}^{-} - L_k \Sigma_{y,k} L_k^T \tag{2.93}$$

  - **SOH** - Update covariance $\Sigma_{\theta,k}^{+}$.

$$\Sigma_{\theta,k}^{+} = \Sigma_{\theta,k}^{-} - L_k^{\theta} \Sigma_{d,k} (L_k^{\theta})^T \tag{2.94}$$

Chapter 3

Results

3.1 OCV-SOC Curve Characterisation

3.1.1 Discharge Test

The discharge test was conducted at a constant target current, selected so that the test duration would be compatible with continuous laboratory supervision, given the nominal capacity of the cell of approximately $30\,Ah$ [3].

$$I_{dis} = \frac{30\,[Ah]}{7.5\,[h]} = 4\,[A] \tag{3.1}$$

The discharge test was conducted with a target current of $4\,A$. The measured current profile and terminal voltage are shown in Figure 3.1. The current remained approximately constant throughout the experiment, with an average value of $3.931\,A$ over a discharge duration of $7.019\,h$. The voltage profile exhibits the characteristic behaviour of a lithium-ion cell, consisting of an initial voltage drop followed by a gradual decrease during most of the discharge process and a steeper decline near the end of discharge.

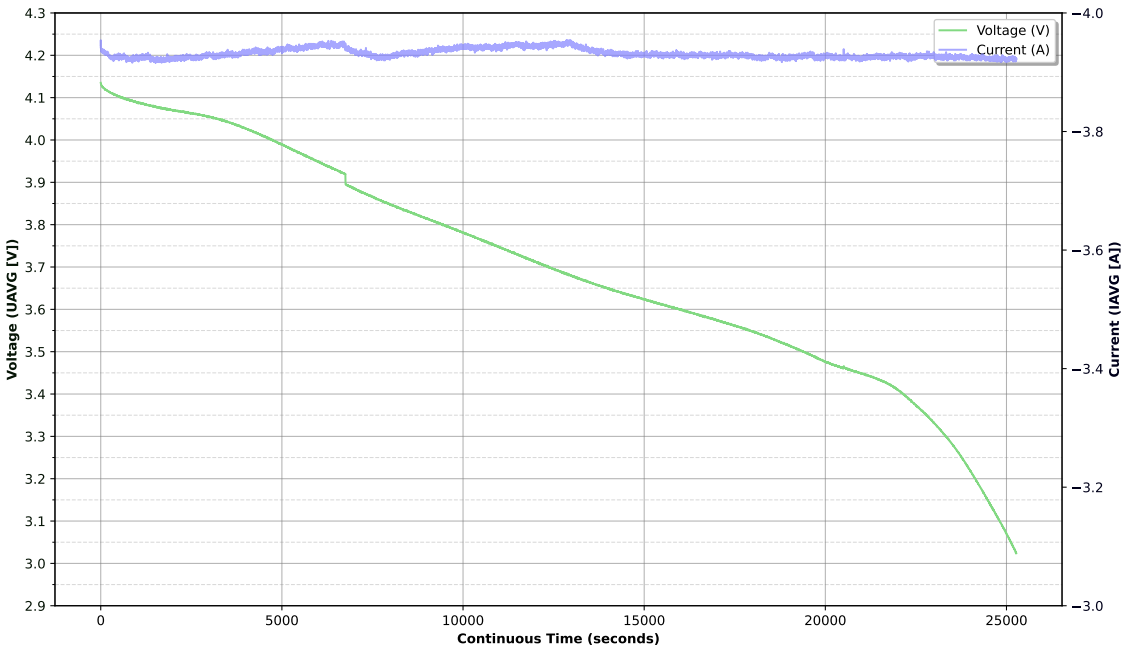


Figure 3.1: Current (lilac) and voltage (green) curves of the NMC battery during the discharge test.

Although the discharge current was regulated to a constant target value, minor deviations were observed due to the control and measurement accuracy of the test equipment. Nevertheless, these variations were sufficiently small to justify the assumption of a constant current for the capacity calculation.
Using the measured average current and the total discharge duration, the actual cell capacity was determined as,

$$Q = (3.931\,[A])(7.019\,[h]) = 27.596\,[Ah] \tag{3.2}$$

The measured capacity of $27.596\,Ah$ is approximately 8.0% lower than the $30\,Ah$ nominal capacity specified by the manufacturer. This difference may be attributed to testing conditions, cell ageing, or the selected discharge cut-off voltage.
The total energy delivered by the cell during the discharge test was obtained from the measured voltage and current data. The power analyser directly provided the accumulated energy in watt-seconds, which was converted to watt-hours according to the following equation,

$$E_{dis} = \frac{366,982.4\,[Ws]}{3,600} = 101.939\,[Wh] \tag{3.3}$$

The measured discharged energy was $101.939\,Wh$, corresponding to approximately 91.8% of the nominal energy rating of $111\,Wh$ specified by the manufacturer.

3.1.2 Charge Test

The charging test was performed with a maximum charging current of $20\,A$ and a total charging duration of $1.94\,h$. The measured voltage and current profiles are presented in Figure 3.2. The results exhibit the expected CC-CV charging behaviour of lithium-ion cells.

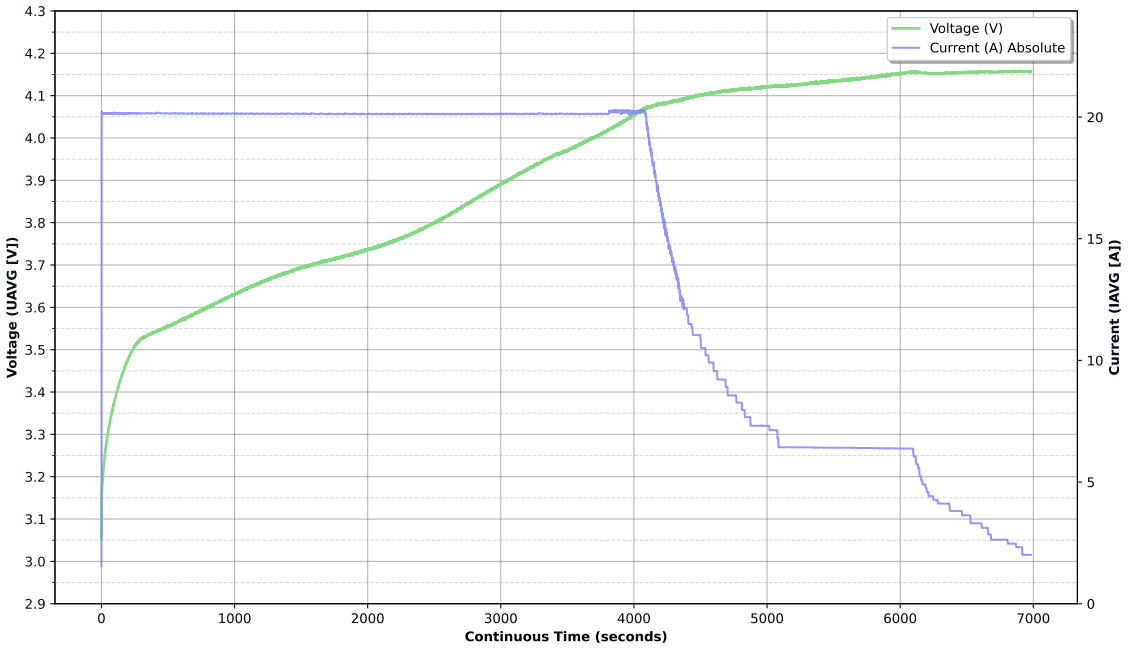


Figure 3.2: Current (lilac) and voltage (green) curves of the NMC battery during the charge test.

During the constant-current phase, the charging current remained close to 20 A while the terminal voltage increased progressively. Once the cell voltage reached approximately 4.05 V, the charger transitioned to the constant-voltage phase, where the charging current gradually decreased and the voltage slowly increased. The charging process was terminated when the voltage reached 4.15 V, current dropped below approximately 2.009 A.

$$E_{charge} = \frac{400,964.76\,[Ws]}{3,600} = 111.379\,[Wh] \tag{3.4}$$

The total energy supplied to the battery during charging was 111.379 Wh. Then, the efficiency was calculated based on the measured this charging and discharging energies, the energy efficiency of the cell was 91.5%.

$$\eta = (100)\frac{101.939\,[Wh]}{111.379\,[Wh]} = 91.524\% \tag{3.5}$$

While the manually calculated capacity from Equation (3.2) was 27.596 Ah, the battery analyser's direct accumulated-Ah reading of 27.616 Ah was adopted as the reference capacity for all subsequent parameter identification and EKF implementation, as the two values differ by less than 0.1%.

3.1.3 Pulses Test

The pulse-discharge test was conducted using SOC intervals of 5%, with a relaxation period of three minutes between consecutive pulses. The pulse magnitude and duration were selected such that each pulse removed an approximately constant amount of charge from the cell.
Using the experimentally determined capacity from Equation (3.2), the charge corresponding to each 5% SOC decrement was calculated as,

$$Q_{step} = (27.616\,[Ah])(5\%) = 1.3808\,[Ah] \tag{3.6}$$

Three discharge current levels were employed during the test. For the first eighteen SOC intervals (approximately 90% of the usable capacity), a current of $10\,A$ was used. As the cell approached the lower voltage region, the discharge current was reduced to $3A$, and finally to $1\,A$ near the end of discharge to improve the resolution of the low SOC measurements.

$$t_{dis_{10A}} = \frac{1.3808\,[Ah]}{10\,[A]}60 \approx 8\,[minutes] \tag{3.7}$$

$$t_{dis_{3A}} = \frac{1.3808\,[Ah]}{3\,[A]}60 \approx 27\,[minutes] \tag{3.8}$$

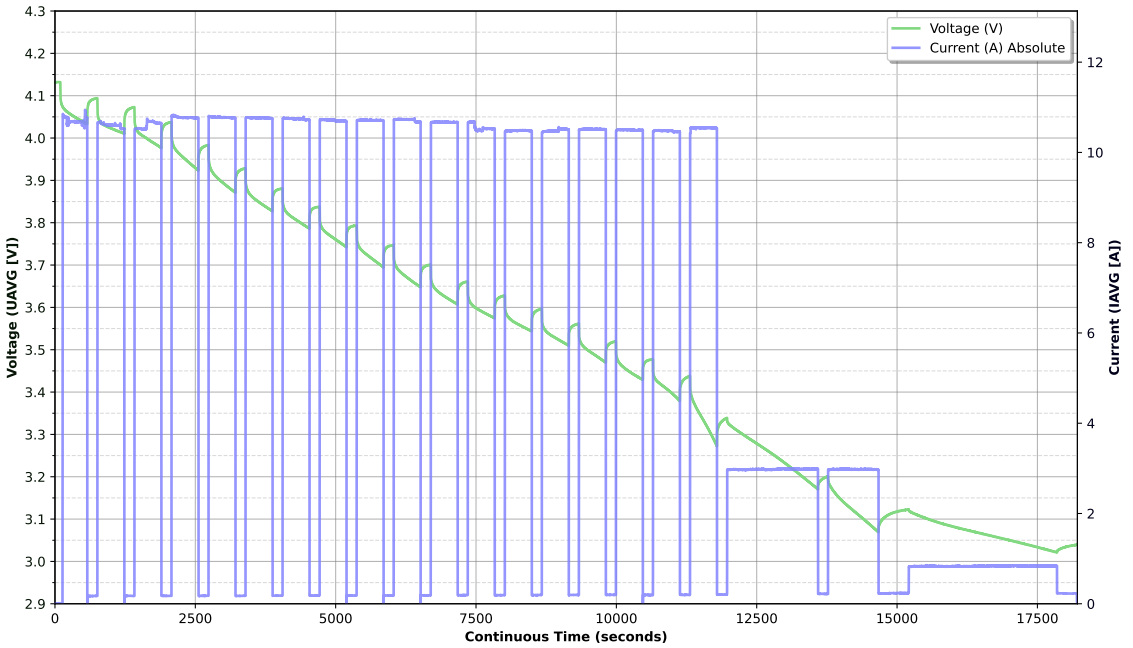


Figure 3.3: Current (lilac) and voltage (green) curves of the NMC battery during the pulses test.

Figure 3.3 presents the complete pulse discharge test. During each current pulse, the terminal voltage decreases due to both the open-circuit voltage variation and the internal voltage drops associated with the cell impedance. Once the current is interrupted, the voltage partially recovers during the relaxation period and gradually approaches an equilibrium value.

The voltage measured at the end of each relaxation period was used as an approximation of the open-circuit voltage corresponding to the current SOC level. These OCV-SOC pairs were subsequently used to construct the OCV-SOC relationship shown in Figure 3.4.

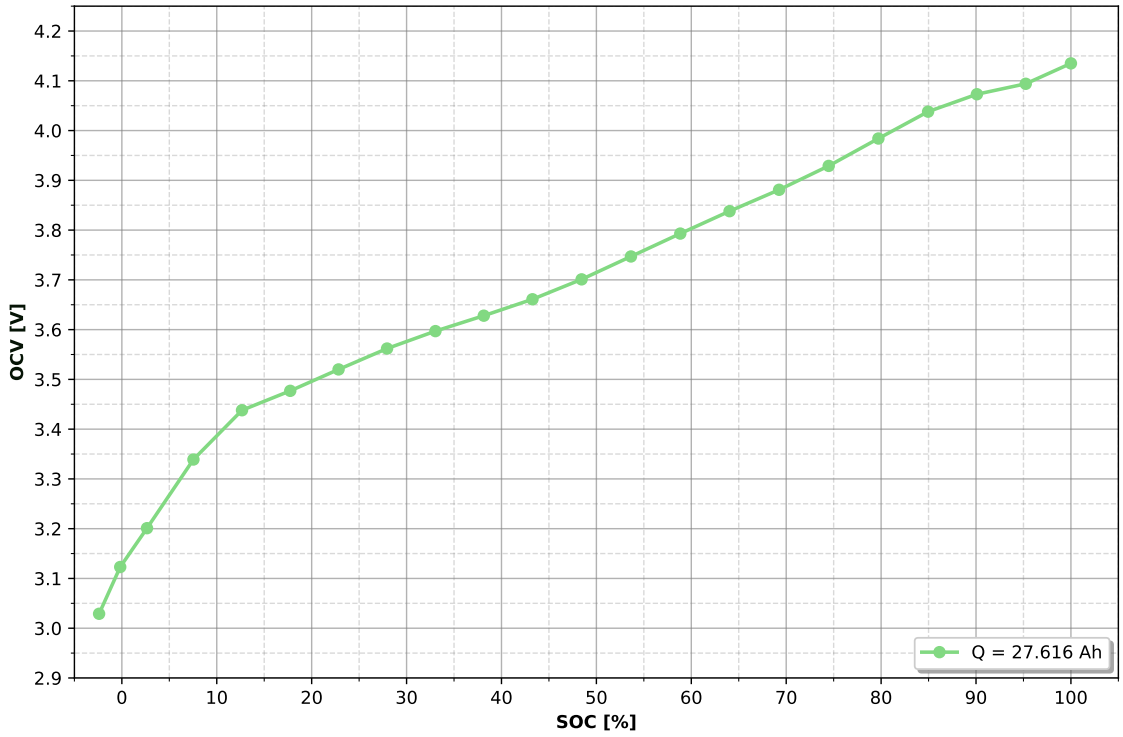


Figure 3.4: OCV-SOC curve of the NMC battery.

The resulting OCV-SOC curve exhibits the characteristic nonlinear behaviour of NMC lithium-ion cells. The voltage varies gradually throughout the middle region, while steeper voltage gradients are observed near fully charged and fully discharged conditions. The measured OCV ranged from $4.138\,V$ at 100% SOC to $3.029\,V$ at 0% SOC.
The experimentally obtained OCV-SOC relationship was subsequently implemented as a lookup table within the equivalent-circuit model and for the EKF estimation algorithm.

3.2 Parameter Identification

3.2.1 R_0, R_1, R_2, C_1, C_2 parameters

The parameter identification test was performed by subjecting the cell to a two-level step-current sequence at each SOC operating point, the first pulse was $10\,A$ for $90\,s$ followed immediately by a second pulse of $30\,A$ for $90\,s$, with a relaxation period of 10 minutes between each consecutive cycle. This protocol was repeated until the battery reached the minimum voltage established as $3.0\,V$.
The algebraic identification method [5] was applied to each cycle independently to extract the six equivalent-circuit parameters (R_0, R_1, R_2, C_1, C_2, and the open-circuit voltage E). To assess the reliability of each identification, the extracted E value was compared against the OCV-SOC curve established from the pulse discharge test shown in Section 3.1.3. Cycles for which the deviation $|E_{\text{extracted}} - \text{OCV(SOC)}|$ exceeded 20 mV were considered numerically unreliable and excluded from the R_1 and C_1 tables. This criterion excluded two cycles and the remaining cycles spanned from 16.6% to 99.9% SOC, which were retained for R_1 and C_1. All twelve cycles were used for R_0, R_2, and C_2, as these parameters showed consistent values across the full range.
The identified resistance values are presented in Figure 3.5. The ohmic resistance R_0 remained approximately constant across the SOC range, with values between $2.1\,\text{m}\Omega$ and $2.6\,\text{m}\Omega$. The diffusion resistance R_2 also showed low variability, remaining below $1.1\,\text{m}\Omega$. The polarisation

resistance R_1 exhibited larger variability, consistent with the known sensitivity of this parameter to the observation window of the algebraic method [5], and is discussed further in Section 7.4.

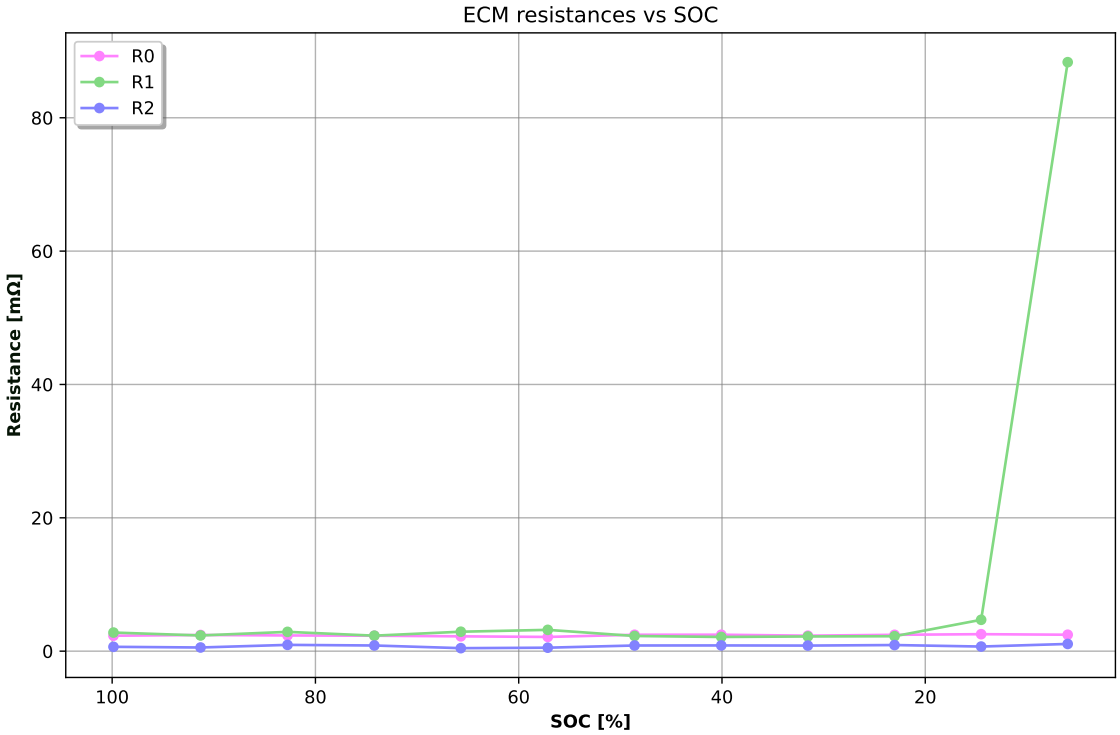


Figure 3.5: Resistors R_0 (pink), R_1 (green), and R_2 (lilac) values of the NMC battery.

The capacitance values, shown in Figure 3.6 on a logarithmic scale, ranged from approximately $18,000\,\mathrm{F}$ to $33,000\,\mathrm{F}$ for C_1 and from $4,000\,\mathrm{F}$ to $13,000\,\mathrm{F}$ for C_2 across the SOC points.

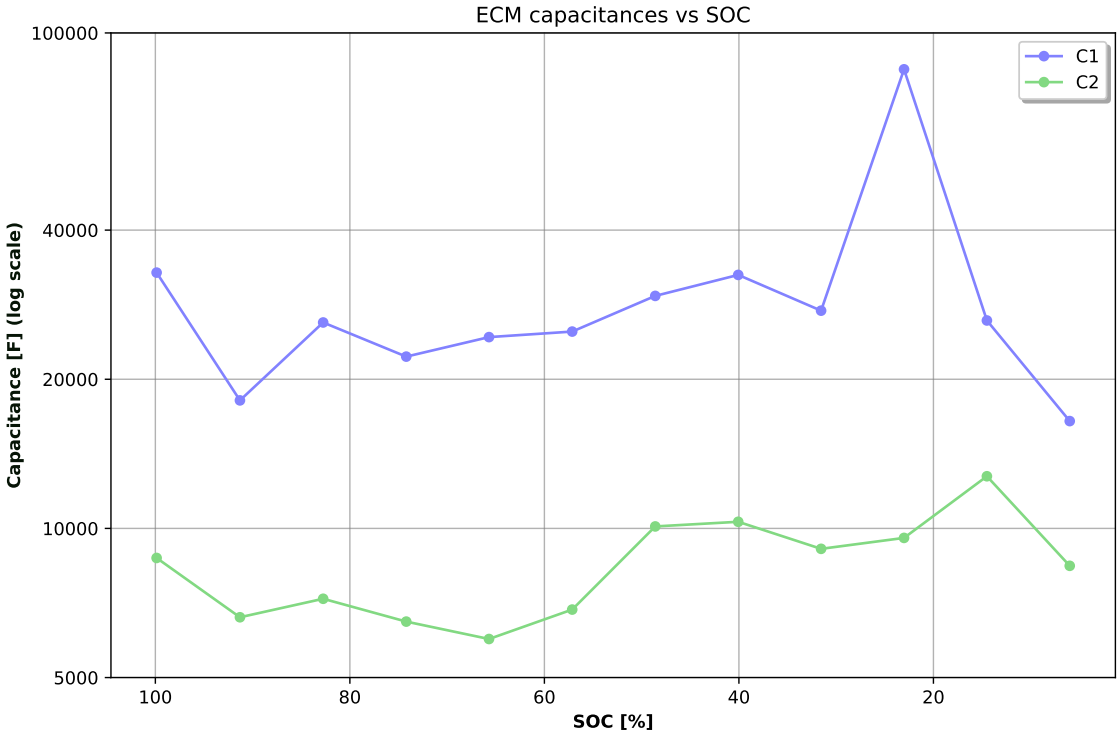


Figure 3.6: Capacitors C_1 (lilac) and C_2 (green) values of the NMC battery.

Figure 3.7 presents a comparison between the OCV–SOC curve obtained from the pulse discharge test and the E values extracted from the parameter identification cycles. The two curves show similar behaviour across the middle SOC range, confirming the consistency of the parameter identification procedure. The mean absolute deviation between the two curves, computed over the ten validated cycles, was below 10 mV.

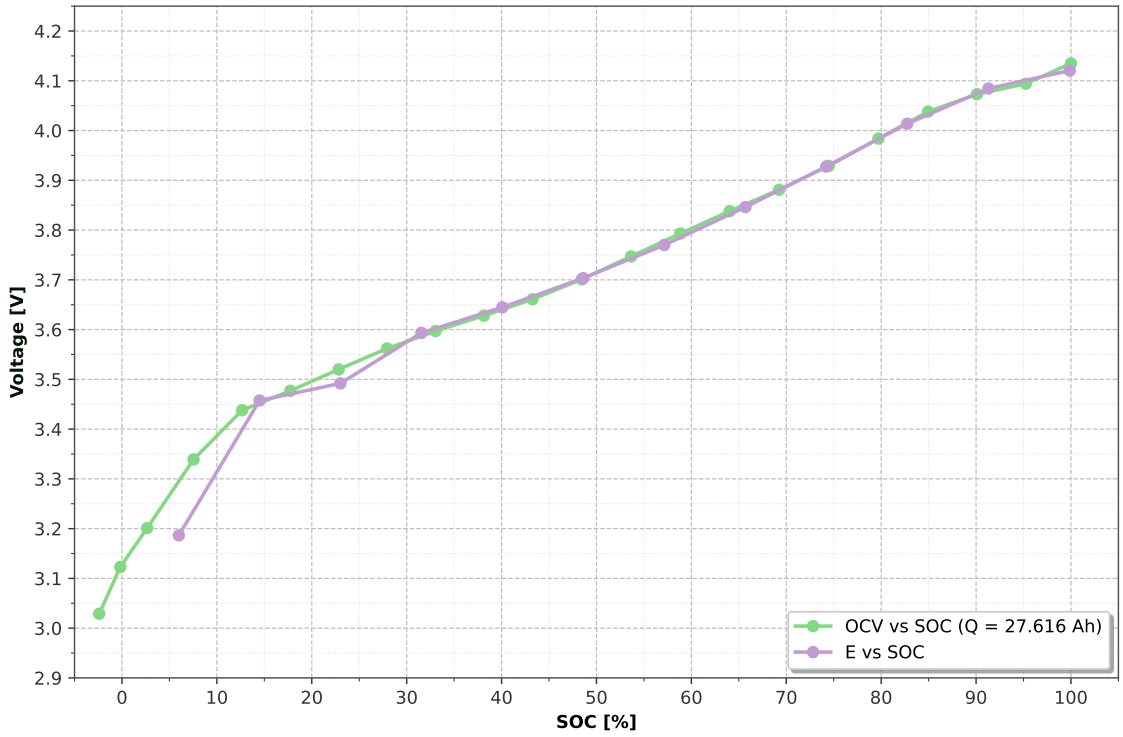


Figure 3.7: OCV vs. SOC curves, obtained through algebraic method (lilac) [5] and curve obtained by pulse test (green) in Section 3.1.3 of the NMC battery

3.3 Extended Kalman Filter

3.3.1 Estimation of SOC with EKF

The SOC EKF estimator was evaluated under three operating profiles: a charging cycle, a cyclic pulse discharge, and a random step-current discharge.

Charge Test The results of the charge test are shown in Figures 3.8 and 3.9. The EKF estimate tracks the Coulomb counting reference closely throughout the 115 min charging cycle, with the error remaining within the 3σ bounds for most of the test. A small divergence is observed in the final phase of charging known as the constant-voltage stage, reducing the observability of the cell state. The Root Mean Square Error (RMSE) between the EKF estimate and the Coulomb counting reference was 0.942%.

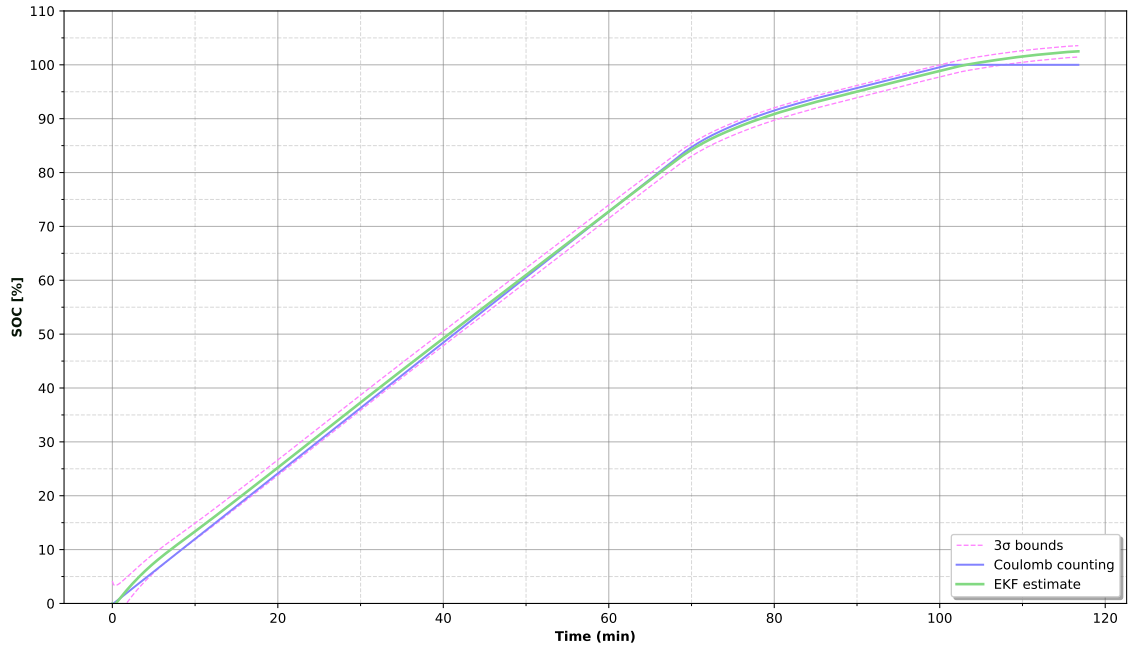


Figure 3.8: SOC in time comparing Coulomb Counting method (lilac) and EKF estimator (green) during the charging cycle of the NMC battery.

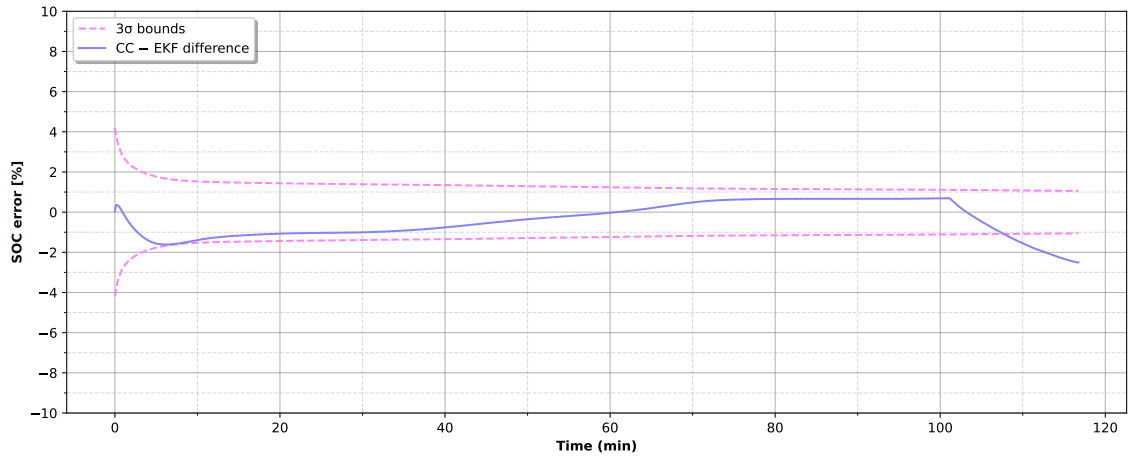


Figure 3.9: Error between CC and EKF SOC estimator during the charging cycle of the NMC battery.

Pulses Test During the cyclic pulse discharge test, the current alternated between 10 A and rest at regular intervals. The EKF estimate is presented in Figures 3.10 and 3.11. The filter tracked the step SOC decrease over the approximately 300 min test duration. The estimation error remained within the 3σ bounds throughout, converging to within $\pm 1\%$ of the Coulomb counting reference after the initial transient. The RMSE obtained was 1.141%.

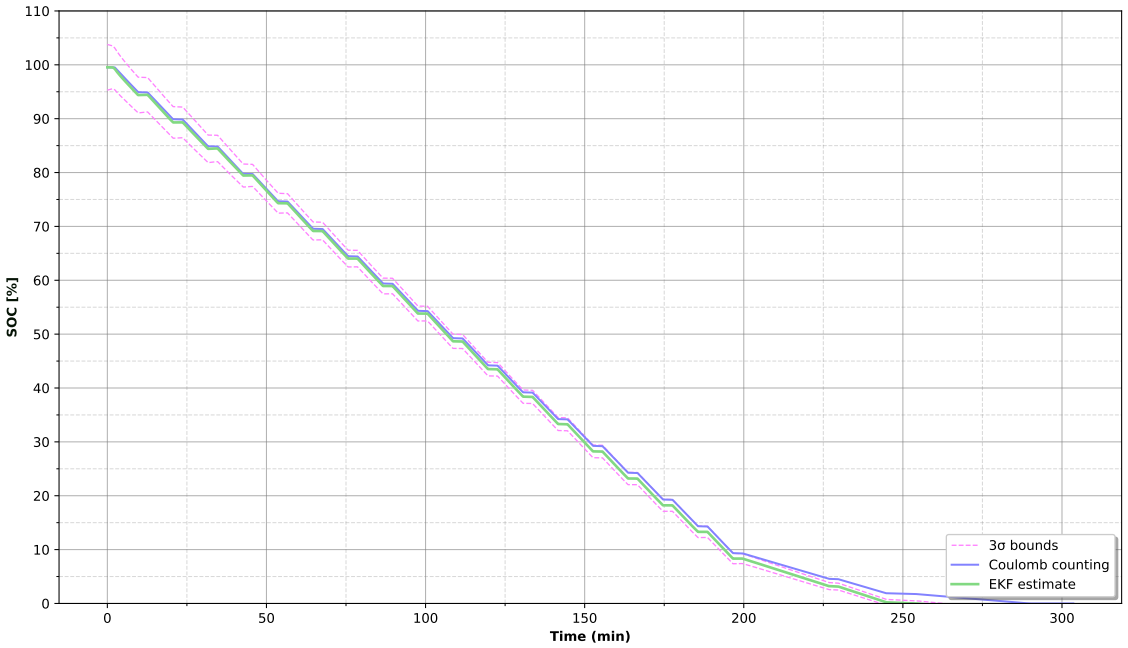


Figure 3.10: SOC in time comparing Coulomb Counting method (lilac) and EKF estimator (green) during the pulses cycle of the NMC battery.

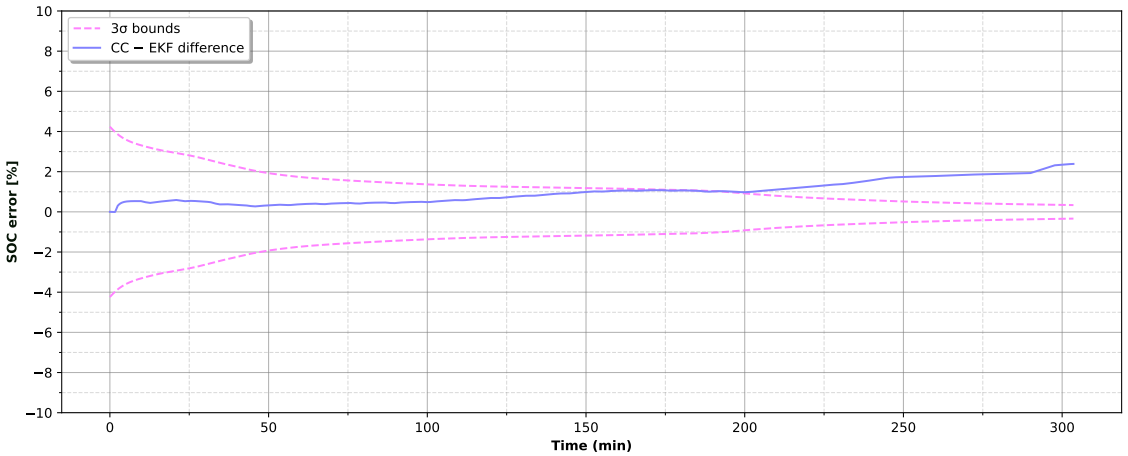


Figure 3.11: Error between CC and EKF SOC estimator during the pulses cycle of the NMC battery.

Random Steps Test The random step-current profile, alternating between 0 A, 10 A, and 30 A, tested the filter's response to abrupt current changes. Results are presented in Figures 3.12 and 3.13. The EKF maintained a similar value to the Coulomb counting reference throughout the test. This scenario produced the lowest RMSE of the three profiles, at 0.691%.

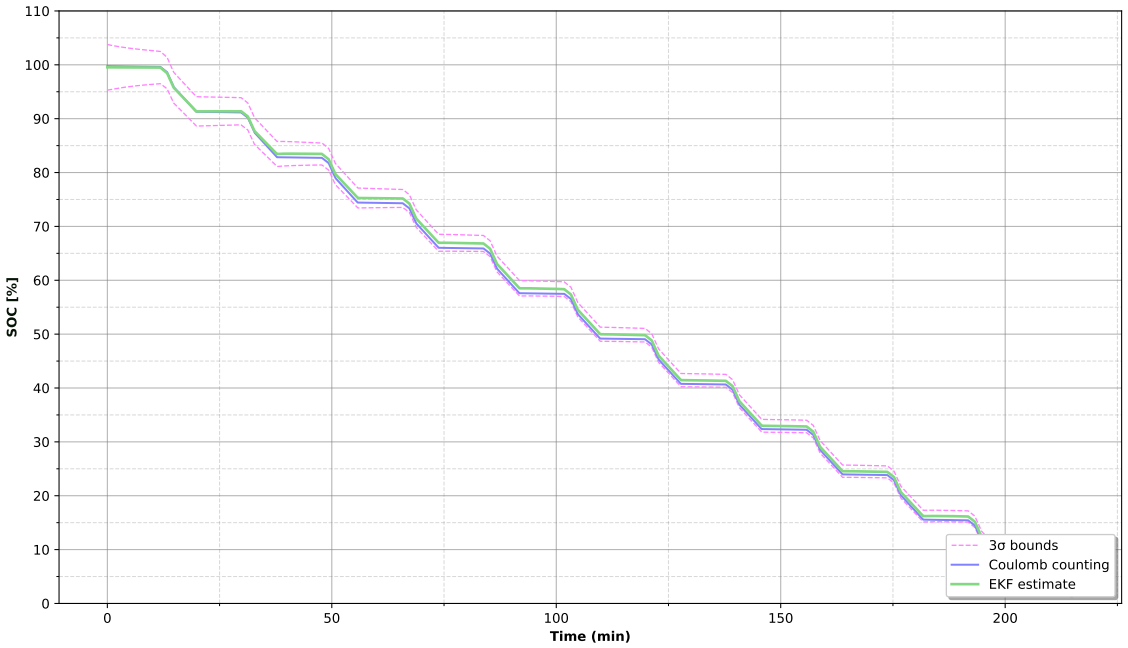


Figure 3.12: SOC in time comparing Coulomb Counting method (lilac) and EKF estimator (green) during the random steps cycle of the NMC battery.

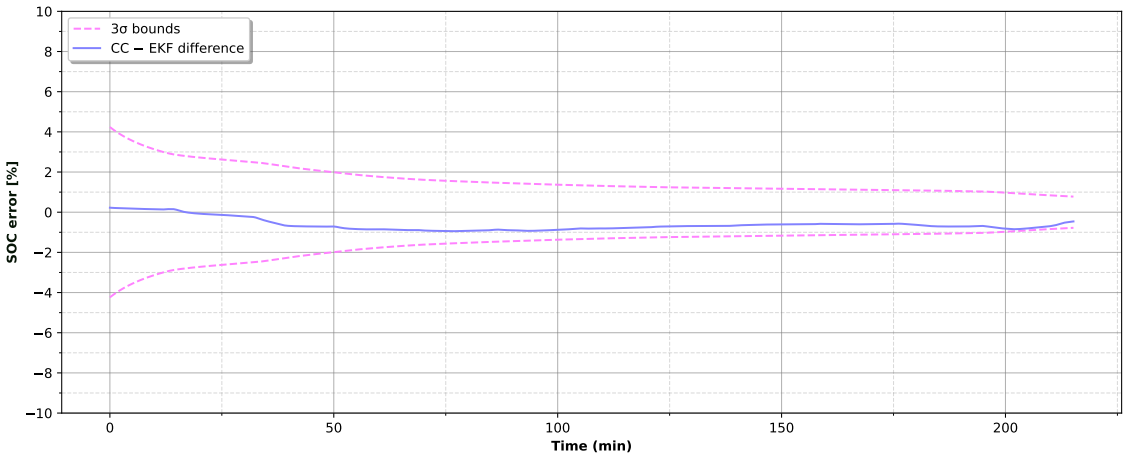


Figure 3.13: Error between CC and EKF SOC estimator during the random steps cycle of the NMC battery.

3.3.2 Estimation of SOC and SOH with Dual EKF

The dual EKF was applied to the same three test profiles, simultaneously estimating the SOC and the SOH, the last state consisting in parameters R_0 and Q. Both parameters were initialised at the values established during cell characterisation. A warm-up period of 500 samples was used, during which the parameter filter accumulated the sensitivity information required for reliable estimation before beginning to update R_0 and Q.

Charge Test The estimated capacity Q and series resistance R_0 are presented in Figures 3.14 and 3.15, respectively. At the end of the test, the capacity estimate had converged to 28.44 Ah,

$3\sigma = \pm 0.58\,\text{Ah}$ and the resistance estimate to $R_0 = 2.246\,\text{m}\Omega$, $3\sigma = \pm 0.462\,\text{m}\Omega$.

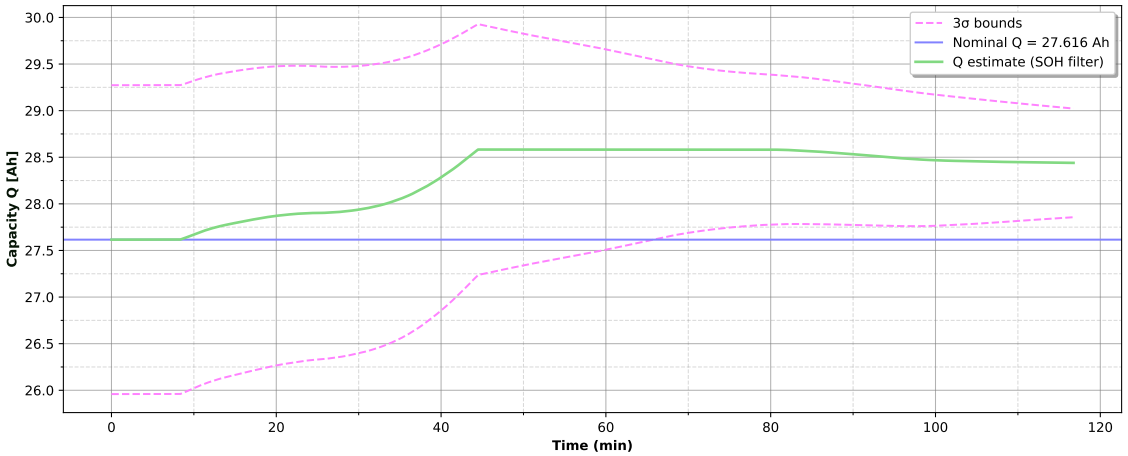


Figure 3.14: Capacity evolution during the CC-CV charging cycle of the NMC battery (constant-current phase at 20 A up to approximately 70 min, followed by the constant-voltage phase until the current dropped below 2.009 A), sampled at $\Delta t = 1\,\text{s}$. Estimated capacity (green) and nominal capacity (lilac).

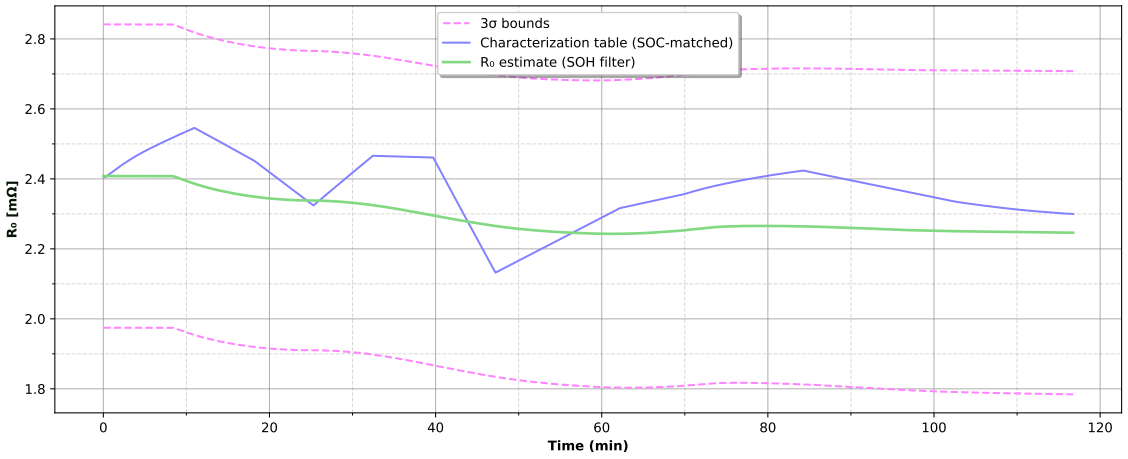


Figure 3.15: Resistor R_0 evolution during the CC-CV charging cycle of the NMC battery (constant-current phase at 20 A up to approximately 70 min, followed by the constant-voltage phase), sampled at $\Delta t = 1\,\text{s}$. Estimated resistor (green) and characterised resistor (lilac).

The SOC estimate for the charge test is shown in Figure 3.16, with the corresponding error in Figure 3.17. The RMSE versus Coulomb counting was 1.390%, higher than the SOC-only estimator result, which is expected since the dual filter simultaneously adapts the capacity estimate.

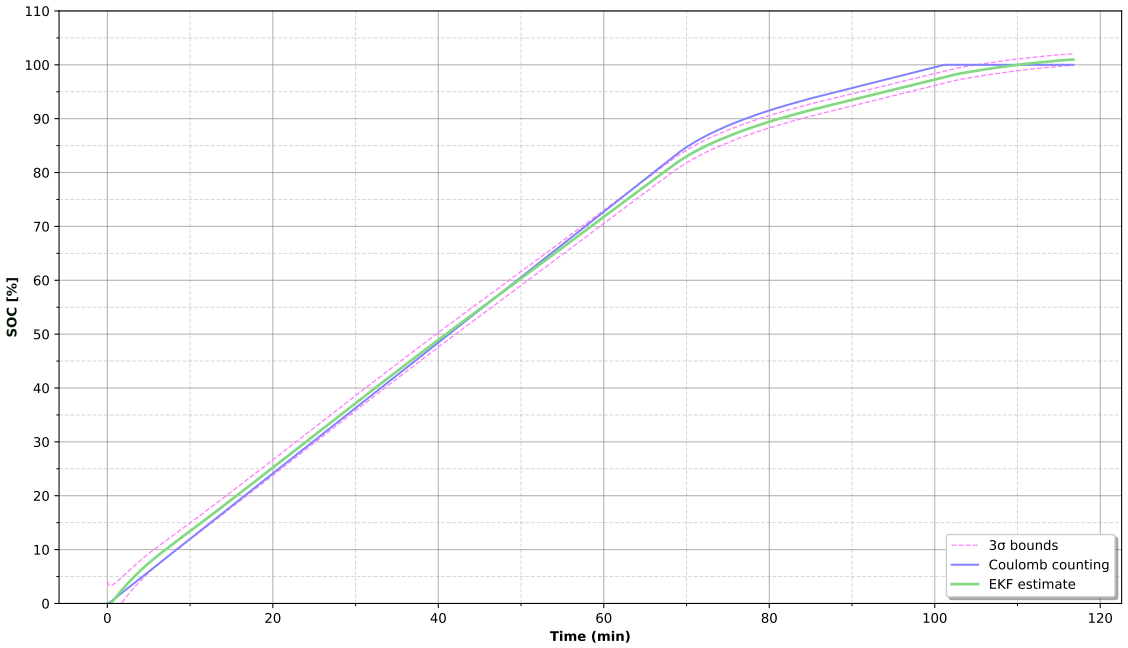


Figure 3.16: SOC in time comparing Coulomb Counting method (lilac) and SOC & SOH EKF estimator (green) during the charging cycle of the NMC battery.

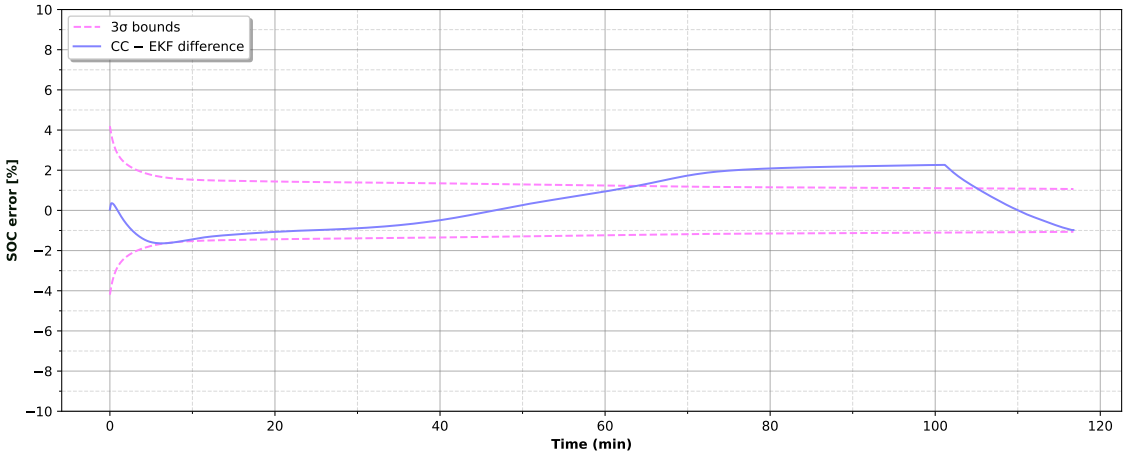


Figure 3.17: Error between CC and dual EKF SOC estimator during the charging cycle of the NMC battery.

Pulses Test The capacity estimate converged to 27.67 Ah, $3\sigma = \pm 0.10$ Ah, closely matching the characterised value. The series resistance converged to 2.134 mΩ, $3\sigma = \pm 0.540$ mΩ. The estimated R_0 remained within the characterisation table range throughout, as shown in Figure 3.19.

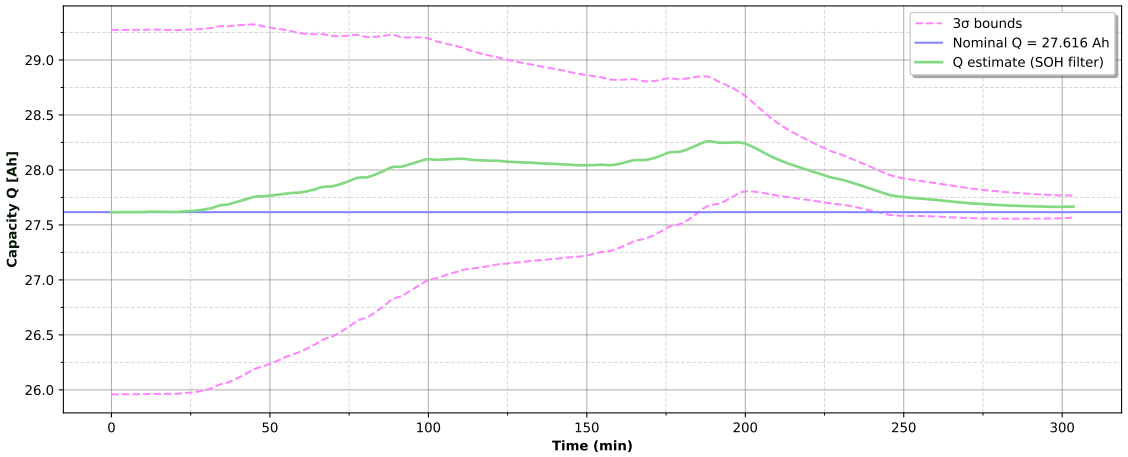


Figure 3.18: Capacity evolution during the cyclic pulse discharge of the NMC battery (10 A pulses of 10 min separated by 3 min rest periods, until full discharge), sampled at $\Delta t = 1\,\mathrm{s}$. Estimated capacity (green) and nominal capacity (lilac).

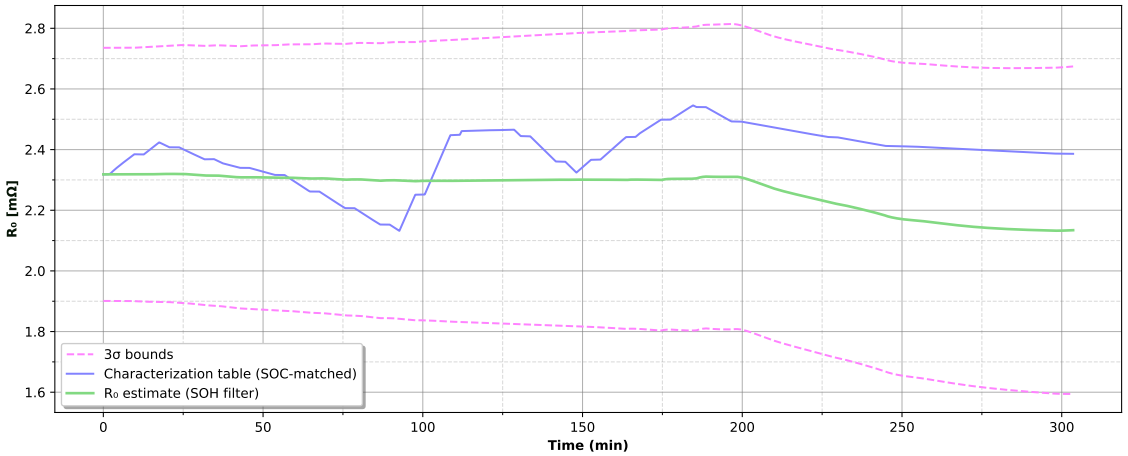


Figure 3.19: Resistor evolution during the cyclic pulse discharge of the NMC battery (10 A pulses of 10 min separated by 3 min rest periods, until full discharge), sampled at $\Delta t = 1\,\mathrm{s}$. Estimated resistor (green) and characterised resistor (lilac).

Results for the cyclic pulse discharge are shown in Figures 3.18–3.21. The SOC RMSE was 0.943%.

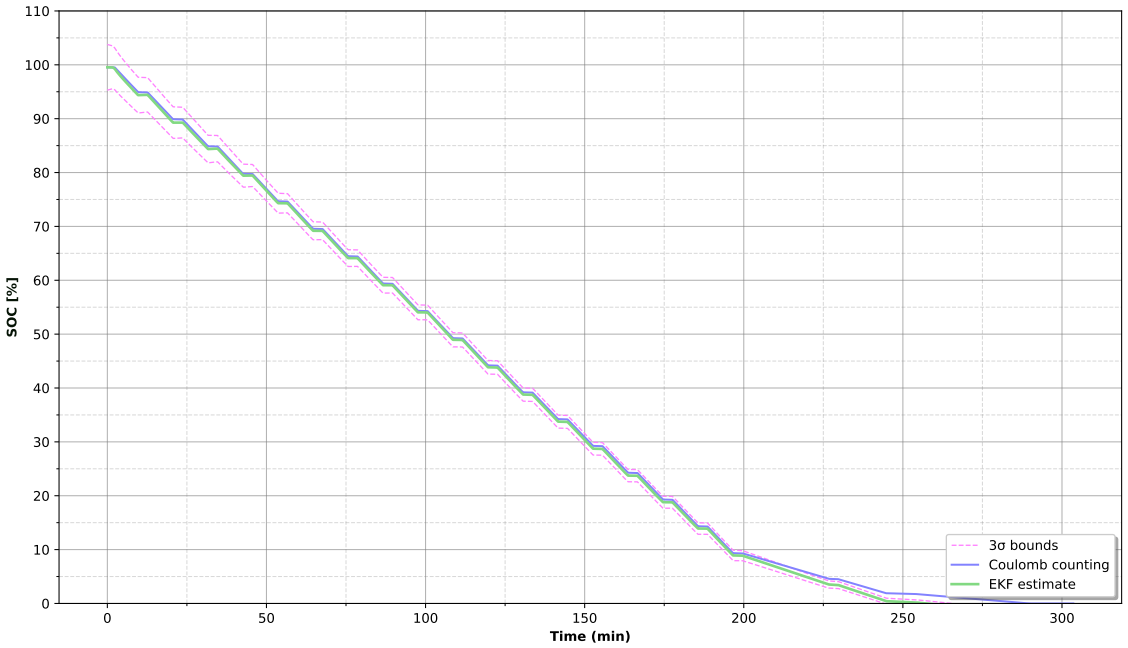


Figure 3.20: SOC in time comparing Coulomb Counting method (lilac) and SOC & SOH EKF estimator (green) during the pulses cycle of the NMC battery.

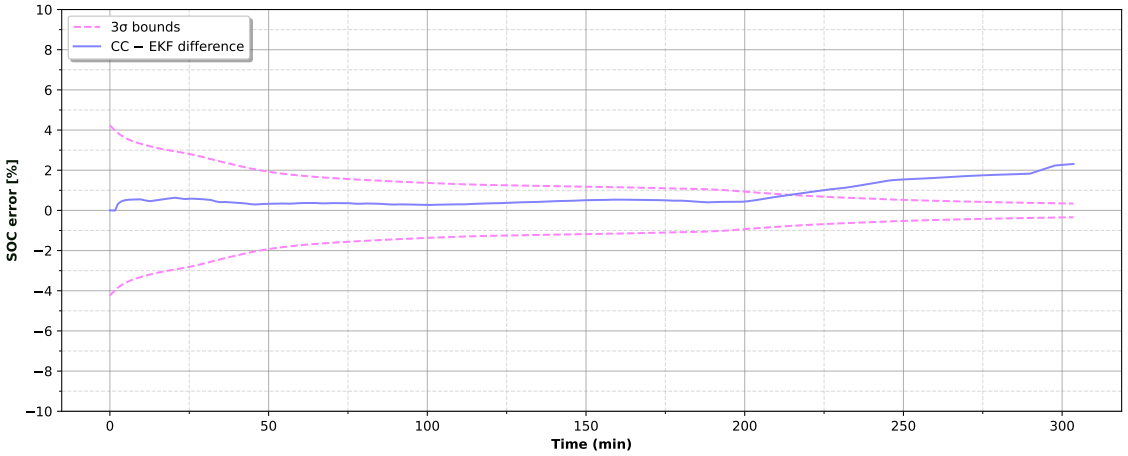


Figure 3.21: Error between CC and dual EKF SOC estimator during the pulses cycle of the NMC battery.

Random Steps Test The final capacity estimate was 27.92 Ah, $3\sigma = \pm 0.32$ Ah and the resistance estimate 2.132 mΩ, $3\sigma = \pm 0.468$ mΩ. The higher current values of this profile resulted in faster initial convergence of R_0 compared to the constant-current tests, as visible in Figure 3.23.

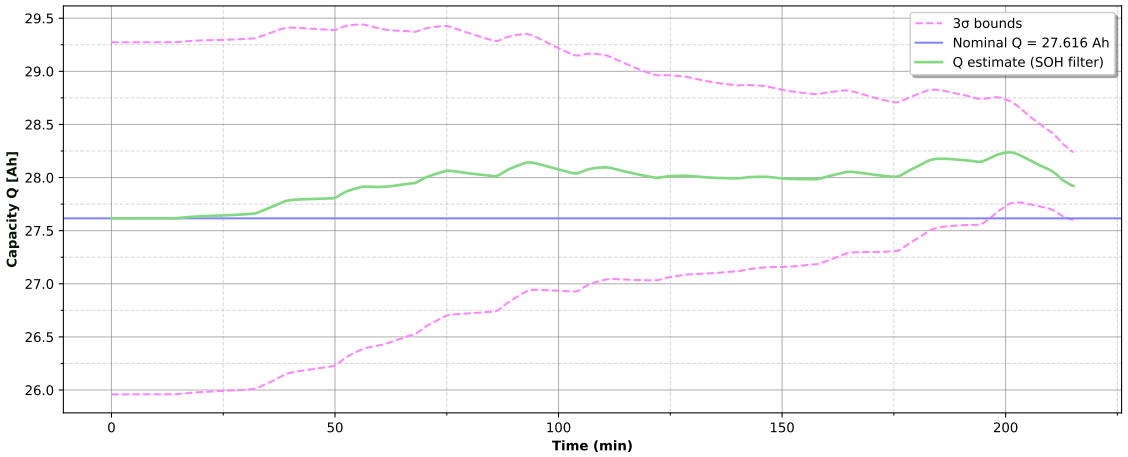


Figure 3.22: apacity evolution during the random step-current discharge of the NMC battery (steps between 0 A, 10 A, and 30 A), sampled at $\Delta t = 1\,\mathrm{s}$. Estimated capacity (green) and nominal capacity (lilac).

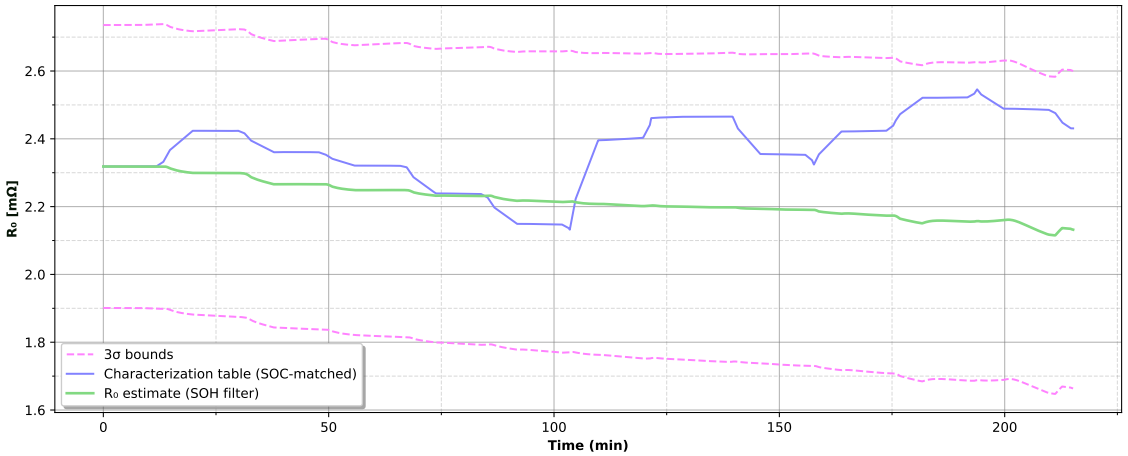


Figure 3.23: Resistor R_0 evolution during the random step-current discharge of the NMC battery (steps between 0 A, 10 A, and 30 A), sampled at $\Delta t = 1\,\mathrm{s}$. Estimated resistor (green) and characterised resistor (lilac).

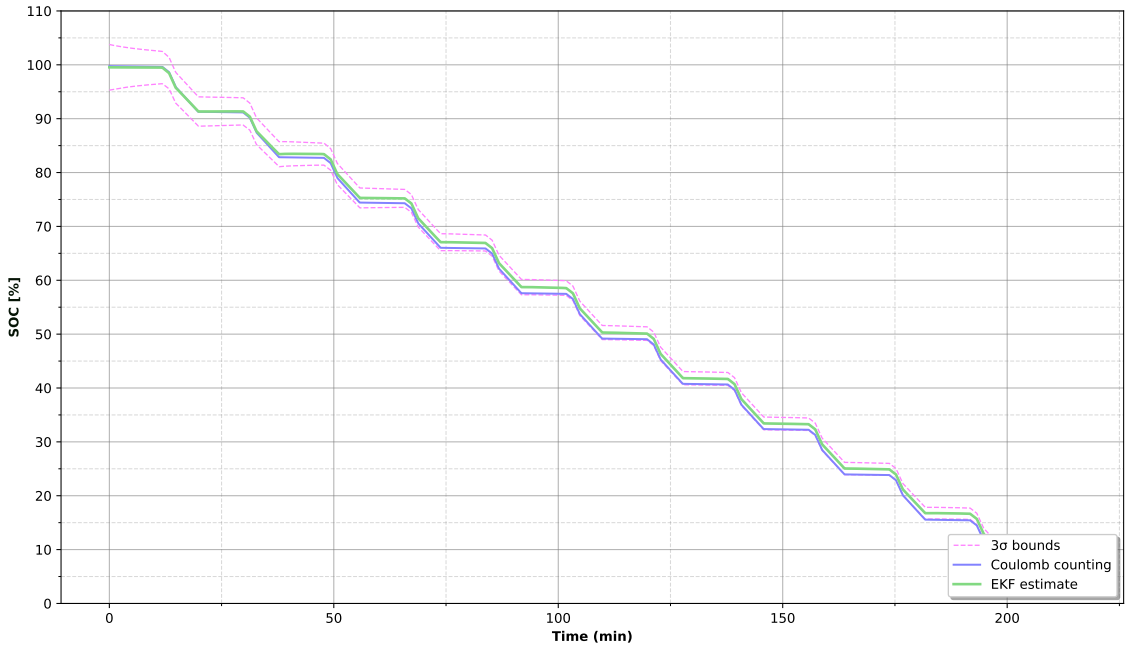


Figure 3.24: SOC in time comparing Coulomb Counting method (lilac) and SOC & SOH EKF estimator (green) during the random steps cycle of the NMC battery.

The random step-current profile produced a SOC RMSE of 0.958% (shown in Figures 3.24 and 3.25).

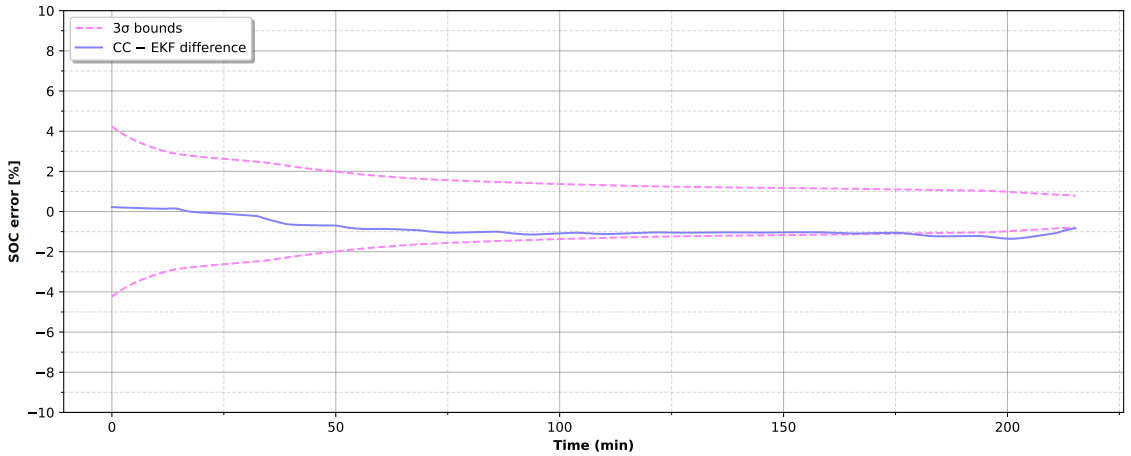


Figure 3.25: Error between CC and dual EKF SOC estimator during the random steps cycle of the NMC battery.

3.3.3 Estimation Comparison between SOC & SOH EKF and SOC filter

Figures 3.26, 3.28, and 3.30 directly compare the SOC filter and dual EKF estimates against the CC reference. Although the dual filter's SOC RMSE was higher (as seen in Figures 3.27, 3.29, and 3.31), both estimators track the reference closely throughout the tests, confirming that the addition of simultaneous SOH estimation does not come at a significant cost to SOC accuracy.

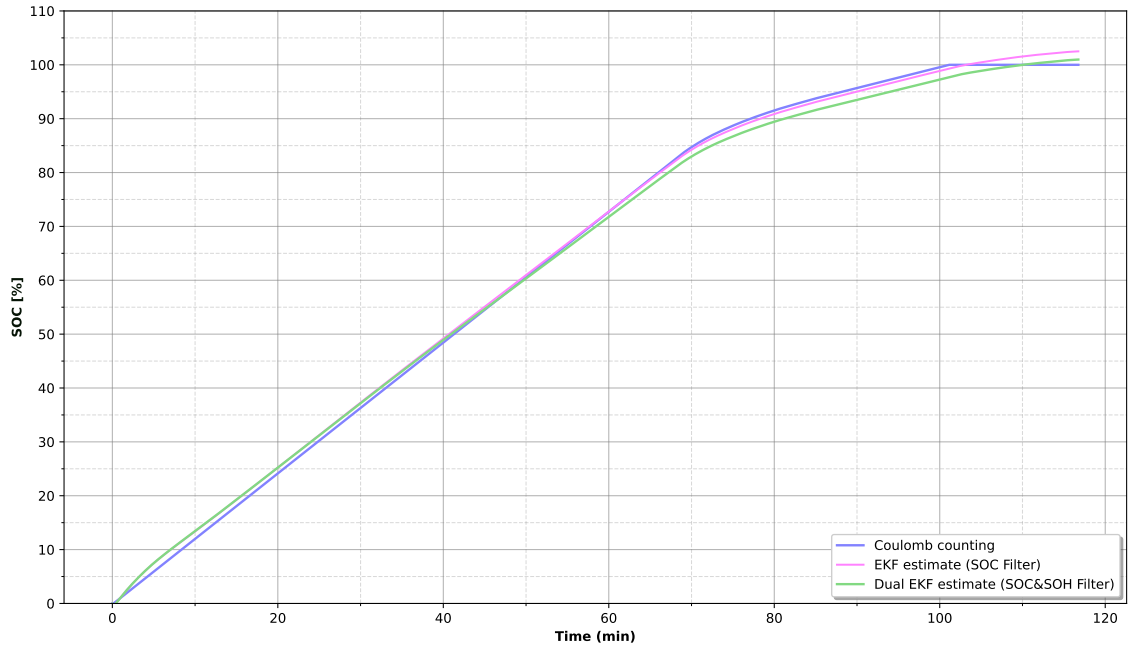


Figure 3.26: Comparison of SOC estimation between the Coulomb-counting reference (lilac), the SOC EKF filter (pink), and the SOC & SOH EKF (green) during the CC-CV charging cycle (constant-current phase at 20 A up to approximately 70 min, followed by the constant-voltage phase), sampled at $\Delta t = 1\,\mathrm{s}$.

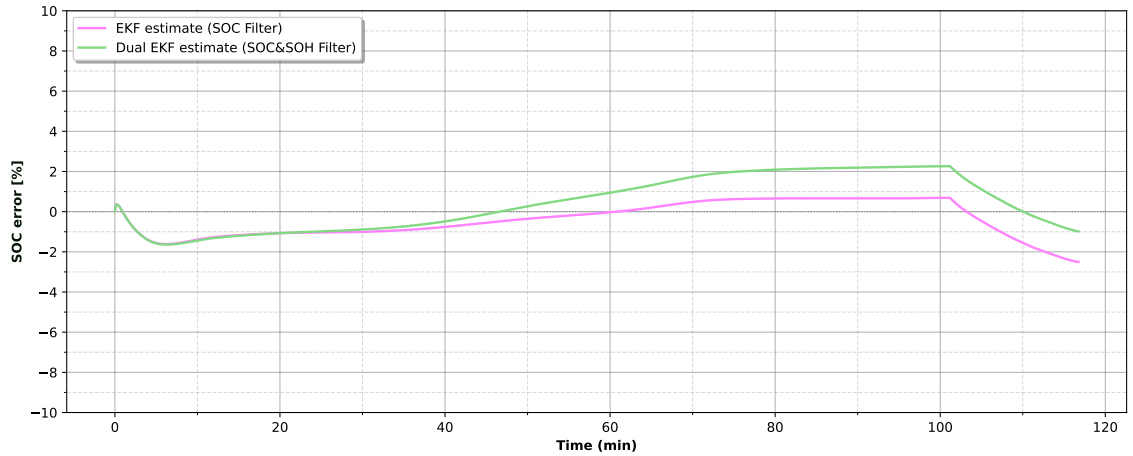


Figure 3.27: Comparison of Error between CC and the SOC EKF filter (pink), and the SOC & SOH EKF (green) during the CC-CV charging cycle (constant-current phase at 20 A up to approximately 70 min, followed by the constant-voltage phase), sampled at $\Delta t = 1\,\mathrm{s}$.

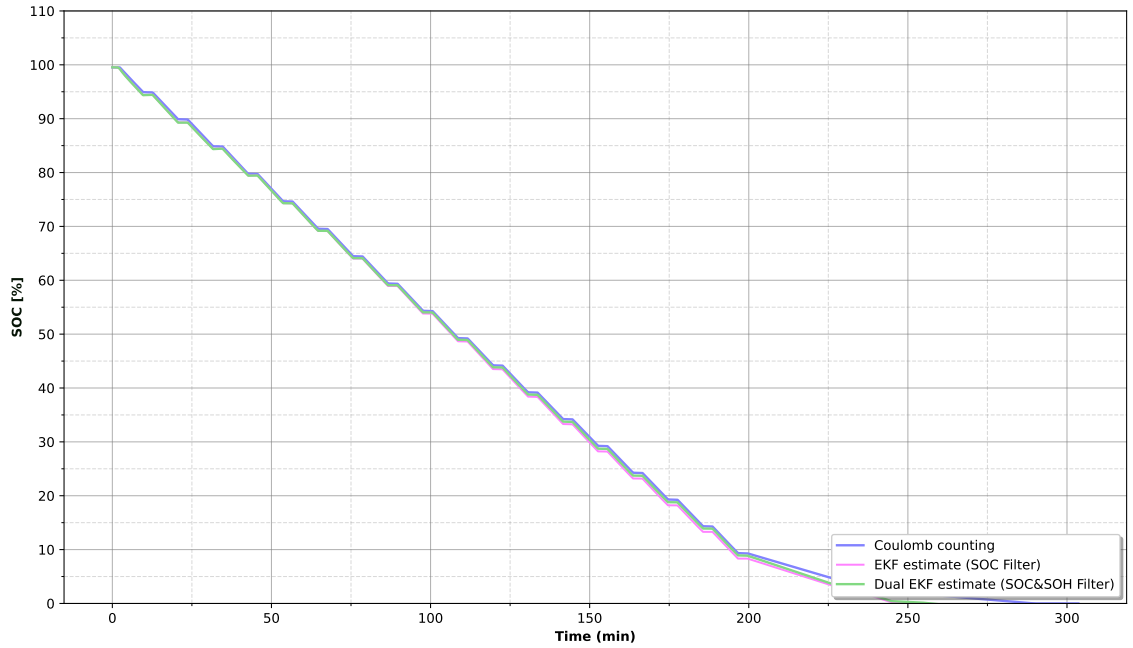


Figure 3.28: Comparison of SOC estimation between the Coulomb-counting reference (lilac), the SOC EKF filter (pink), and the SOC & SOH EKF (green) during the cyclic pulse discharge of the NMC battery (10 A pulses of 10 min separated by 3 min rest periods, until full discharge), sampled at $\Delta t = 1\,\mathrm{s}$.

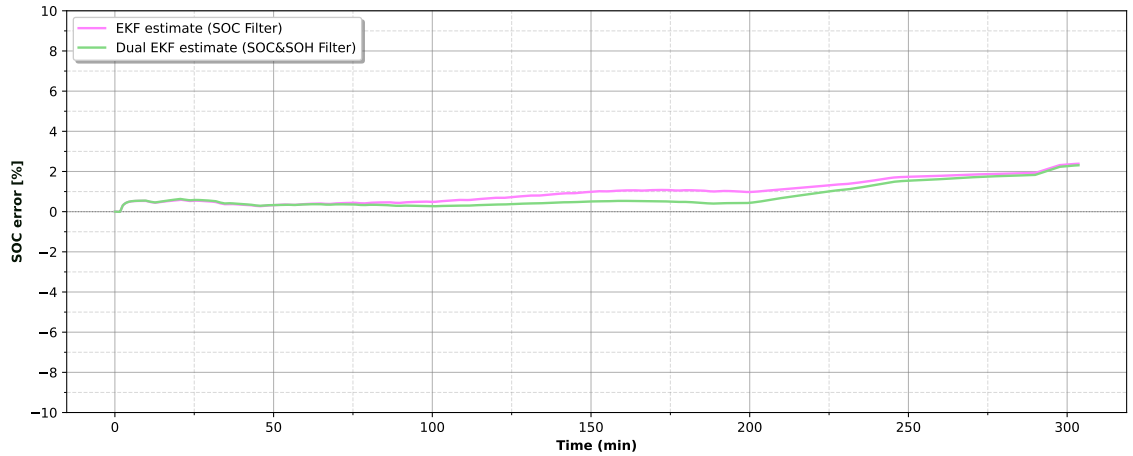


Figure 3.29: Error between CC and the SOC EKF filter (pink), and the SOC & SOH EKF (green) during the cyclic pulse discharge of the NMC battery (10 A pulses of 10 min separated by 3 min rest periods, until full discharge), sampled at $\Delta t = 1\,\mathrm{s}$.

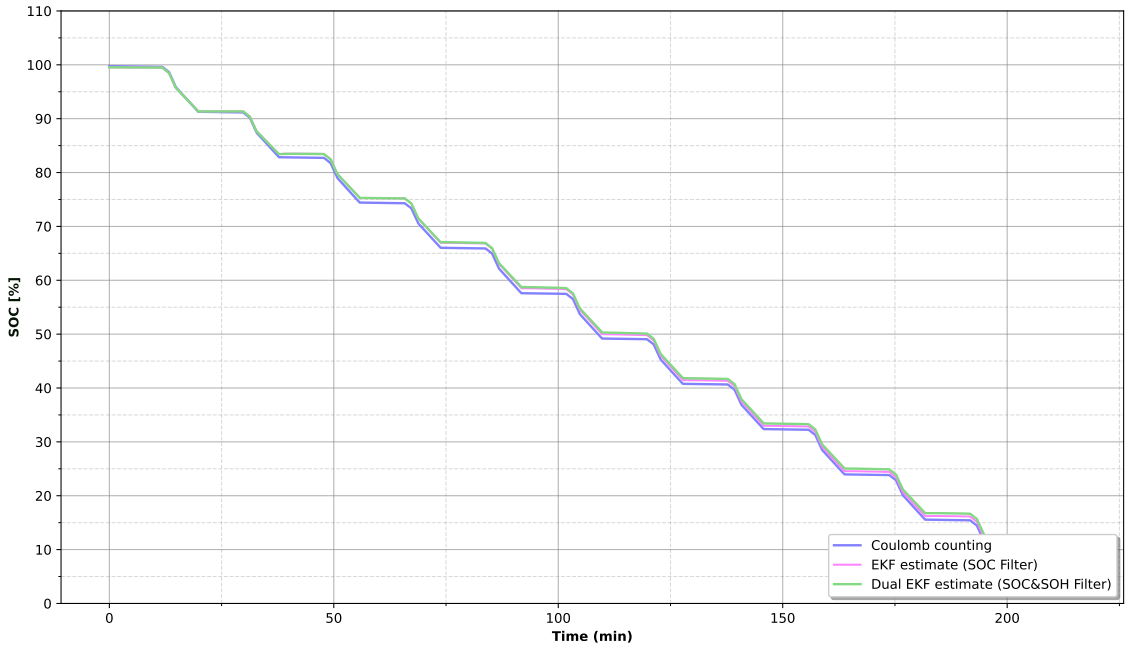


Figure 3.30: Comparison of SOC estimation between the Coulomb-counting reference (lilac), the SOC EKF filter (pink), and the SOC & SOH EKF (green) during the random step-current discharge of the NMC battery (steps between 0 A, 10 A, and 30 A), sampled at $\Delta t = 1\,\mathrm{s}$.

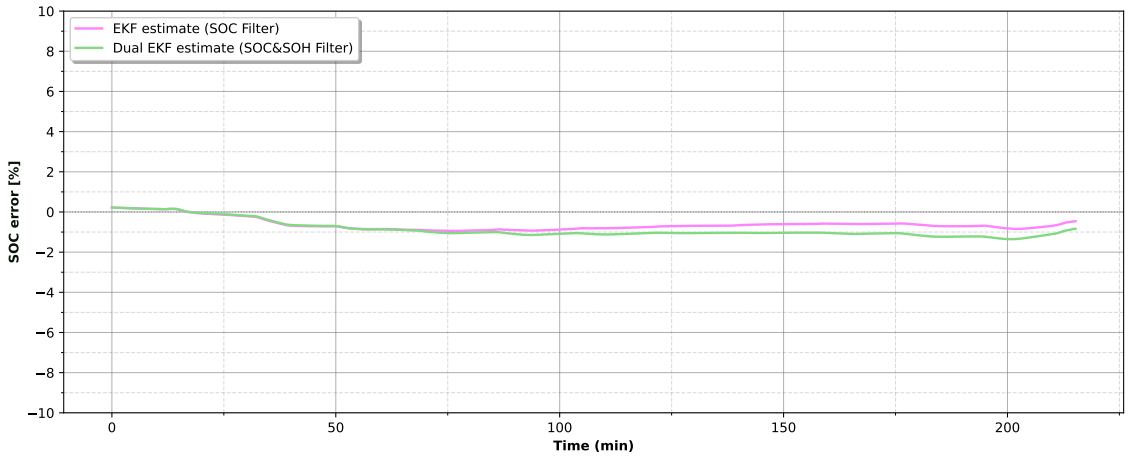
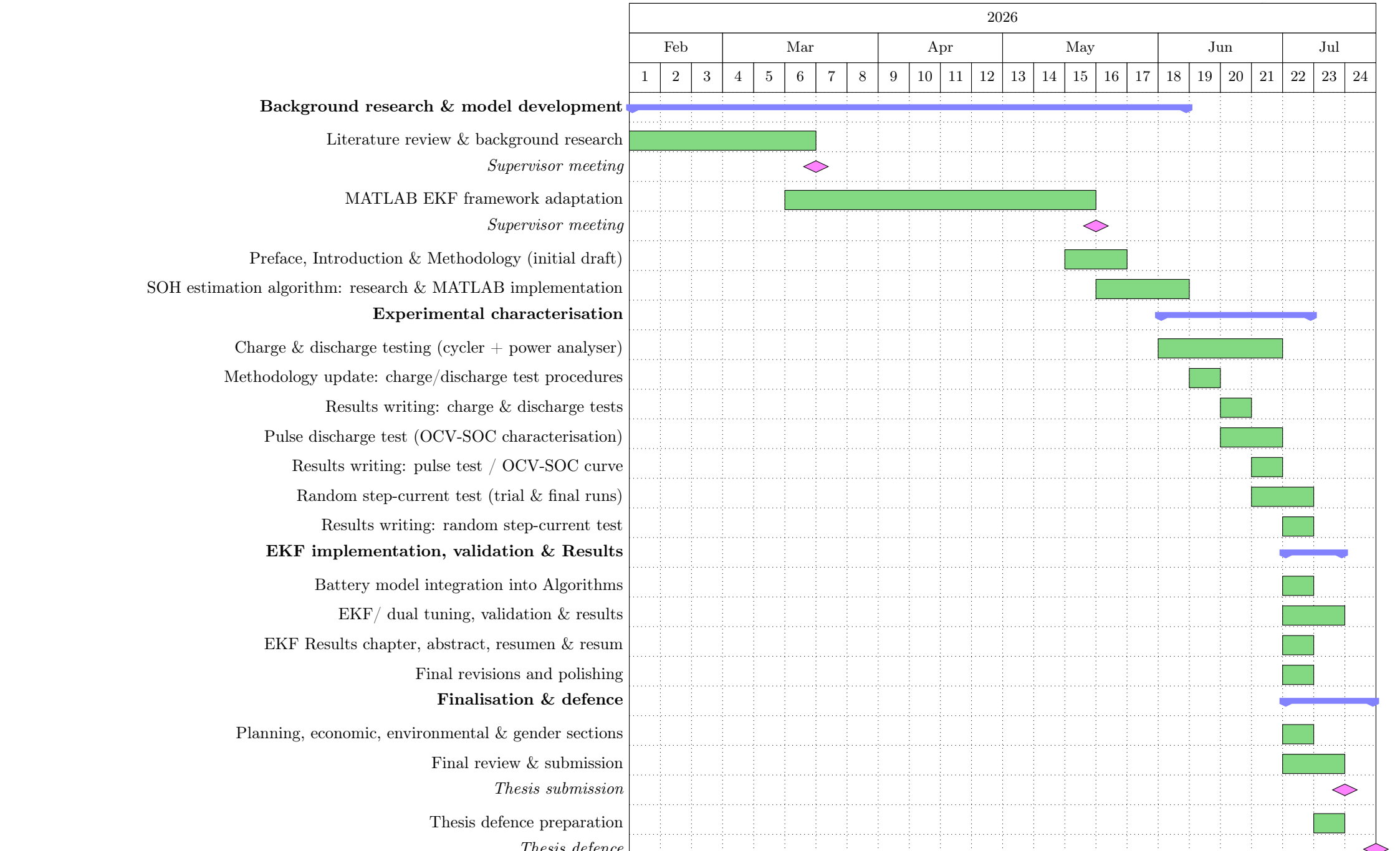


Figure 3.31: Error between CC and the SOC EKF filter (pink), and the SOC & SOH EKF (green) during the random step-current discharge of the NMC battery (steps between 0 A, 10 A, and 30 A), sampled at $\Delta t = 1\,\mathrm{s}$.

Chapter 4

Planning

The work presented in this thesis was carried out between February and July 2026, divided into four phases: literature review and model development, experimental characterisation of the cell, implementation and validation of the Extended Kalman Filter algorithms, and thesis writing. Figure 4.1 presents the Gantt chart summarising the timeline of the work performed.
The characterisation phase was constrained by the availability of laboratory equipment (cycler, power analyser, and setup), which delayed the start of the discharge, charge, pulse, and steps tests relative to the initial plan. To keep the project on schedule, the parameter identification code, the SOH estimation algorithm, and the corresponding sections of the Methodology chapter were developed in parallel while waiting for equipment availability and the setup.


2026
Feb
Mar
Apr
May
Jun
Jul
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24
Background research & model development
Literature review & background research
Supervisor meeting
MATLAB EKF framework adaptation
Supervisor meeting
Preface, Introduction & Methodology (initial draft)
SOH estimation algorithm: research & MATLAB implementation
Experimental characterisation
Charge & discharge testing (cycler + power analyser)
Methodology update: charge/discharge test procedures
Results writing: charge & discharge tests
Pulse discharge test (OCV-SOC characterisation)
Results writing: pulse test / OCV-SOC curve
Random step-current test (trial & final runs)
Results writing: random step-current test
EKF implementation, validation & Results
Battery model integration into Algorithms
EKF/ dual tuning, validation & results
EKF Results chapter, abstract, resumen & resum
Final revisions and polishing
Finalisation & defence
Planning, economic, environmental & gender sections
Final review & submission
Thesis submission
Thesis defence preparation
Thesis defence

Chapter 5

Economic Assessment

This chapter presents an economic analysis of the resources needed in carrying out this Master's thesis, considering the student's working hours, operational expenses, and experimental materials. All monetary values are expressed in euros.

5.1 Labour Cost

This thesis corresponds to 30 European Credit Transfer and Accumulation System (ECTS) credits. As established in Article 4.5 of Spanish Royal Decree [14], one ECTS credit represents between 25 and 30 hours of student workload; the lower bound of 25 hours/ECTS is adopted here, giving a total estimated workload of $30 \times 25 = 750$ hours. Following the assessment guidelines provided for the elaboration of this section, the student's time is valued at a nominal rate of 15 /h, even though no wage was paid. Table 5.1 distributes the 750 hours across the main project phases, proportionally to the duration of each phase in the project timeline presented in Figure 4.1.

Phase	**Hours**	**Cost [Euros]**
Literature review & background research	187.5	2,812.50
MATLAB software EKF framework adaptation	281.3	4,218.75
Methodology drafting & SOH algorithm development	93.8	1,406.25
Experimental testing & associated results writing	125.0	1,875.00
Final EKF integration, results & thesis completion	62.5	937.50
Total	**750.0**	**11,250.00**

Table 5.1: Labour cost breakdown by project phase, at 15 *Euros*/h.

5.2 Operational Expenses

The cell was recharged approximately 20 times over the course of the project before most test sessions, drawing approximately 111.4 Wh per recharge, as measured directly during the charge test (Equation 3.4). The power analyser and current clamp were estimated to be in active use for approximately 40 hours in total, at an assumed power of 20 W.

Source	Energy [kWh]
Cell recharging (20 cycles × 111.4 Wh)	2.228
Laboratory instruments (40 h × 20 W)	0.800
Total	**3.028**

Table 5.2: Estimated electricity consumption.

Using the 2026 annual average regulated electricity price of 0.128 *Euro*/kWh [2], the total electricity cost was:

$$3.028\,\text{kWh} \times 0.128\,Euro/\text{kWh} = 0.387\,Euro \tag{5.1}$$

5.3 Experimental Materials

The cell characterisation setup required nine high-power resistors, used as the discharge load bank for all discharge tests, at an approximate unit cost of 4.93 *Euro* each, provided by the eRacing team:

$$9 \times 4.93\,Euro = 44.37\,Euro \tag{5.2}$$

The test setup also included a Texas Instruments microcontroller and six power inductors, which were provided by the eRacing team as part of the existing test bench, and two Welion SHP270–30 cells, also provided by the team. The individual purchase costs of these items were not available at the time of writing and are therefore not itemised in this work.

5.4 Software Licensing

MATLAB was used throughout this work for algorithm implementation and simulation. Although the licence used was provided by the university, its economic value is estimated using the listed MathWorks Student annual subscription for 2026 85 *Euro*/year [1], prorated for approximately 5.3 months during which the software was actively used for this work:

$$85\,Euro/\text{year} \times \frac{5.3}{12} = 37.54\,Euro \tag{5.3}$$

5.5 Total Cost

Table 5.3 summarises the total estimated cost of this thesis.

Item	Cost [€]
Student labour (750 h × 15 €/h)	11,250.00
Electricity	0.39
Experimental materials (resistors)	44.37
MATLAB! (**MATLAB!**) licence (prorated)	37.54
Total (excl. VAT)	**11,332.30**

Table 5.3: Total estimated cost of the thesis.

The total cost of the work amounts to 11,332.30 *Euro*.

5.6 Economic Assessment Discussion

The labour cost accounts for approximately 98% of the estimated budget. The majority of the effort was intellectual and computational (literature review, algorithm development, and thesis writing) rather than reliant on equipment or materials, since the experimental setup used cells, instrumentation, and a test bench already available to the eRacing team and the university, at minimal marginal cost to this specific project.

Chapter 6

Environmental Assessment

This chapter assesses the environmental impact of carrying out this Master's thesis, expressed in terms of greenhouse gas emissions, Carbon Dioxide Equivalent (CO_2e), resulting from the electricity consumed during the experimental and computational work.

6.1 Electricity Consumption

Three sources of electricity consumption were considered: the energy drawn from the grid to recharge the cell, the operating power of the laboratory instruments during the tests, and the power consumed by the student's laptop throughout the project.

Cell recharging. The cell was fully recharged approximately 15 times over the course of the characterisation and validation tests, drawing approximately 111.4 Wh per charge, as measured directly during the charge test (Equation 3.4):

$$15 \times 111.4\,\text{Wh} = 1.671\,\text{kWh} \tag{6.1}$$

Laboratory instruments. The power analyser, cycler, and current clamp were in active use for a total of approximately 70.5 hours, distributed across the four test as summarised in Table 6.1, at an assumed operating power of 20 W:

$$70.5\,\text{h} \times 20\,\text{W} = 1.410\,\text{kWh} \tag{6.2}$$

Test campaign	**Hours**
Charge tests (15 sessions × 2 h)	30.0
Discharge tests (2 sessions × 7 h)	14.0
Pulse discharge tests (2 sessions × 5 h)	10.0
Random step-current tests (2 full × 6 h, 3 trial × 1.5 h)	16.5
Total	**70.5**

Table 6.1: Estimated instrument usage hours by test campaign.

Personal computer. The student's laptop was used for approximately 500 hours over the course of the project MATLAB simulations for data processing, and thesis writing at an average power draw of 30 W. Approximately two weeks were spent working from home during the literature

review phase, with the remaining time split approximately 30% at public libraries and 70% in the laboratory.

$$500\,\text{h} \times 30\,\text{W} = 15.000\,\text{kWh} \tag{6.3}$$

Source	**Energy [kWh]**
Cell recharging	1.671
Laboratory instruments	1.410
Personal laptop	15.000
Total	**18.081**

Table 6.2: Total estimated electricity consumption.

6.2 Greenhouse Gas Emissions

The Spanish national electricity grid mix emission factor for 2025, is $0.258\,\text{kg}\,\text{CO}_2e/\text{kWh}$, consistent with the emission factors presented by the Catalan Office for Climate Change [16]. Applying this factor to the total electricity consumption:

$$18.081\,\text{kWh} \times 0.258\,\text{kg}\,\text{CO}_2e/\text{kWh} = 4.665\,\text{kg}\,\text{CO}_2e \tag{6.4}$$

6.3 Environmental Impact Discussion

The estimated carbon footprint of this thesis, $4.665\,\text{kg}\,\text{CO}_2e$. The personal laptop accounts for approximately 83% of the total electricity consumption. The cell recharging and instrument operation together account for less than 20% of the total footprint, reflecting the efficiency of the algebraic parameter identification method used in this work (Section 2.3), which requires substantially shorter tests, and therefore less instrument energy consumption, than conventional characterisation approaches requiring long relaxation periods.

Chapter 7

Social and Gender Equality Assessment

7.1 Access to the Technology

The battery state estimation framework developed in this thesis is not a product or service with differentiated access. It does not discriminate between people of different gender identities, or against vulnerable groups, the algorithms, once published, are equally accessible to any engineer or researcher regardless of gender.

The access to electric vehicle technology, of which battery management is a component, remains unevenly distributed across socio-economic groups due to the current cost of electric vehicles. This is a systemic issue of the electric mobility sector.

7.2 Differentiated Impacts of the Research

The results of this thesis, SOC and SOH estimation accuracy for a racing motorcycle battery, do not have any known differentiated impact on any gender, either directly or indirectly.

7.3 Gender Distribution in Cited Sources

This section examines the gender distribution among the authors of the works cited throughout this document, as a reflection of representation within the research field.

This analysis was performed by inferring the likely gender associated with each author's given name. This method has clear limitations: it can misclassify individuals whose name does not conventionally align with their gender identity, and many given names, particularly transliterated names common in East Asian publications, do not carry a strong gender association for a non-native reader and were therefore classified as ambiguous rather than guessed. Organisations, standards bodies, and manufacturer datasheets cited without an individual named author were excluded from this count, as were entries not ultimately used in this thesis.

Of the 51 individually named authors across the references cited in this thesis, 24 (47.1%) were classified as having a male-associated given name, 4 (7.8%) as female-associated, and 23 (45.1%) as ambiguous. Restricting the analysis to only the 28 names with a reasonably inferable gender, 85.7% are male-associated and 14.3% are female-associated.

This reflects a research field in which women remain a clear minority among identifiable authors, independent of any specific choice made during the literature review process.

7.4 Relevant Sustainable Development Goals

This work contributes to the following Sustainable Development Goals (SDG):

- **SDG 7 – Affordable and Clean Energy**: accurate battery state estimation improves the efficiency and safe utilisation of Li-ion battery capacity, supporting the broader adoption of electric mobility.
- **SDG 9 – Industry, Innovation and Infrastructure**: this work contributes methodological innovation in BMS technology, an enabling infrastructure for electric vehicles.
- **SDG 13 – Climate Action**: by supporting the reliability and performance of electric vehicles, this work contributes indirectly to the decarbonisation of the transport sector, as discussed in the Preface (Section).

Conclusion

This work developed and validated a model-based dual estimation framework for the simultaneous estimation of the SOC and SOH of the WeLion SHP270-30 NMC cell used in the eRacing electric motorcycle at ETSEIB. The framework combines experimental cell characterisation, algebraic parameter identification, and dual EKF implementation in MATLAB. The following sections present the conclusions from each stage of the work.

7.5 Battery Characterisation

The discharge test confirmed an actual cell capacity of 27.616Ah, approximately 8% below the nominal rating of 30Ah. This difference is consistent with typical behaviour in practice, where the rated nominal capacity represents an ideal cell. Factors during the discharge test such as the selected maximum and minimum voltage, conservative test conditions and ageing contributed to this difference.
The charge test confirmed the expected CC-CV behaviour of the Li-ion battery and established an energy efficiency of 91.5%, consistent with values reported for Li-ion cells in the literature [18]. As expected, the Ah measured during charging exceeded the discharge capacity due to the inefficiencies; for this reason, the discharge test capacity was adopted as the reference for all SOC calculations and simulations throughout this work.
The pulse discharge test produced an OCV–SOC curve spanning from 4.138 V corresponding to a 100% SOC to 3.029 V for 0% SOC. The characteristic nonlinear shape, with steeper slope at the higher and lower values of the SOC and an almost linear region in between, is consistent with the typical behaviour of a NMC lithium-ion cells [8]. The resulting lookup table was successfully implemented as the OCV(z) function in the equivalent-circuit model and in both the SOC and SOH EKF algorithms.

7.6 Parameter Identification

The algebraic parameter identification method of Devarakonda and Hu [5] was applied to a two-level circuit. Twelve identification cycles were completed, covering a SOC range from approximately 8% to 99%.
The ohmic resistance R_0 and diffusion resistance R_2 were robustly identified across the full SOC range, showing low variability between cycles. Their values, in the range of 2.1–2.6 mΩ and below 1.1 mΩ respectively, are consistent with expected values for a cell of this capacity [12].
The resistance R_1 and capacitance C_1 exhibited larger variability throughout the cycles. therefore two cycles were excluded by following the criteria ($|E_{\text{extracted}} - \text{OCV(SOC)}| > 20\,\text{mV}$), and the validated cycles covered the range of 16.6%–99.9% SOC. This behaviour is consistent with the sensitivity analysis described in Devarakonda and Hu [5], when the test phase duration is short relative to the slow RC time constant, the algorithm cannot reliably separate the contributions of R_1 and C_1 from the initial capacitor voltage, making the individual values unreliable even when the overall voltage fit is reasonable.
The second-order Thévenin Equivalent Circuit Model (ECM) captures the dominant voltage response of the cell adequately for the purpose of state estimation, as confirmed by the EKF vali-

dation results. However, the model cannot fully represent the complex electrochemical processes ocurring in a cell. In particular, the steep regions of the OCV–SOC curve, the fully charged and fully discharged phases. The complex behaviour in these regions cannot fully be captured by the linear RC structure. This is reflected in the numerical unreliability of R_1 and C_1 at low SOC (below approximately below 16%), and represents a known limitation of the ECM approach at the extremes of the operating range.

7.7 SOC Estimation with EKF

The SOC estimator was evaluated across three current profiles representative of different scenarios. Table 7.1 summarises the RMSE results with respect to the Coulomb-counting reference.

Test profile	**RMSE [%]**
Random steps (0 A, 10 A, 30 A)	0.691
Charge cycle (up to 20 A)	0.942
Cyclic pulses (10 A and rest)	1.141

Table 7.1: SOC estimation RMSE for the EKF SOC filter across the three test profiles.

All three profiles produced RMSE values below 1.15%, and the estimation error remained within the 3σ bounds throughout each test. These results are consistent with values reported in comparable EKF–ECM implementations in the literature, where errors on the order of 1%–2% are typical for similar profiles [13][7].
The random step-current profile produced the lowest RMSE, The charge test produced an intermediate error.

7.8 Simultaneous SOC and SOH Estimation with Dual EKF

The dual EKF was applied to the same three profiles, simultaneously estimating SOC and the SOH parameters R_0 and Q. Table 7.2 summarises the SOC RMSE and the final SOH estimates for each test.

Profile	**SOC RMSE [%]**	Q [Ah]	R_0 [mΩ]
Random steps	0.958	27.92	2.132
Charge cycle	1.390	28.44	2.246
Cyclic pulses	0.943	27.67	2.134

Table 7.2: Dual EKF results: SOC RMSE and final SOH estimates across the three test profiles.

The capacity estimate Q converged to values close to the characterised reference of 27.616 Ah for the discharge profiles, with the tightest convergence observed for the cyclic pulse test 27.67 Ah, 3σ. The charge cycle produced a slightly higher estimate (28.44 Ah), which reflects the Coulombic inefficiency of the charging process. The total energy transferred during a full charge exceeds the energy provided in a discharge cycle, and therefore the parameter filter partially interprets additional charge as a larger capacity.
The resistance estimates R_0 in all three profiles converged to values consistent with those in the characterisation table at the corresponding SOC levels. Values of R_0 exceeding 100% indicate that the online estimate is lower than the nominal table value, reflecting that the filter adapts to the actual operating resistance, which can differ from the characterisation value due to differences in current magnitude, temperature, and operating history between the two tests.

Overall, these results demonstrate that the dual EKF framework is capable of tracking both SOC and the SOH parameters simultaneously and consistently across different current profiles.

7.9 Limitations and Future Work

The equivalent-circuit model does not account for temperature effects nor hysterisis. All experiments were conducted at approximately ambient temperature but in a racing scenario, the cell temperature can vary significantly during a race, which affects all model parameters. Incorporating temperature-dependent parameter tables would improve the accuracy and robustness of the estimator under real operating conditions.
The parameter identification at low and high SOC is incomplete due to the numerical sensitivity of the algebraic method in regions where the OCV gradient is steep, a methodology accounting for the electrochemical behaviour in these regions would improve the results.
Future work could also explore the Unscented Kalman Filter (UKF) as an alternative to the EKF. The UKF avoids the first-order linearisation used by the EKF, and may provide improved accuracy in the nonlinear regions of the OCV–SOC curve without a significant increase in computational cost [17].
Finally, validation under actual race current profiles, including the high peak currents and rapid transients characteristic of a motorcycle competition, would further confirm the applicability of the framework to the operational conditions for which it was designed.

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Institute for Dynamic Systems and Control
Prof. Dr. R. D'Andrea, Prof. Dr. L. Guzzella

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Prof. Dr. Francisco Díaz González
Prof. Dr. Joan Gabriel Bergas Jané

Student:

Name: María José Cuéllar Mejía
E-mail: maria.jose.cuellar@estudiantat.upc.edu
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